

Human Thought

Its Forms and Its Tasks

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Foreword

The life of thought, which it is the task of my book to describe, is intimately interwoven with the other sides of the life of the spirit. I have sought to show that thought cannot serve the life of the spirit as a whole unless it first and foremost becomes master in its own domain and unfolds itself according to its own laws as an independent side of the life of the spirit. That this unfolding, when it is rightly carried through, will bring with it great and far-reaching changes in personal life as a whole, is something of which I become more and more convinced. There are here oppositions and problems that will take on ever sharper forms, and I feel no admiration for those who here sense no sting, although they have sufficient opportunity to come to know the life of thought.

Just as I have on the whole stressed the personal factor in all thinking, the more this approaches the border-regions, so too I have not concealed that my own stance at thought's final station has a personal character, and I have not been able to refrain from stating how, in my experience, one feels at this station.

I still have much to learn. My work will need filling out and correction on many points. But beyond the framework I have given here, I cannot go. And it affords me a good place to work. May there be one or another who could also find work here, even if this work should lead him to alter both the framework and its content.

August 1910.

Harald Høffding.

I. The Psychology of Thought (Functions)

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A. Psychic Energy

1. The task I have set myself in this book is to investigate human thought—a task that all philosophers have set themselves, whether they have been conscious of it or not. For all questions that can be raised concerning life and existence depend—both with respect to the way in which they are posed and with respect to the way in which they are answered—on the conception one has of the nature and operation of thought. It is therefore also an outline of my philosophy that I shall give here. That is to say: I shall give an account of the thoughts to which I have come back again and again, when I have occupied myself with such questions as: the presuppositions of science, the significance of the results of science for one's view of life, and the relation between scientific cognition and other sides of the human life of the spirit.

2. We know thought best as reflection. The word *Eftertanke*—"after-thought"—I use not only in its strict sense, "in one's thinking to *seek after* something or *pursue* something," but also in the sense that thought *comes after* an immediate experience. Reflection thus presupposes that we have seen, felt, experienced something. During the experience we were perhaps wholly absorbed in it; but now we seek to bring it forth anew, either to dwell once more on what was experienced, or to go through its various parts or properties or its relations to other experiences. In mere dwelling, reflection is one with a beholding, a mere representing. But such dwelling can scarcely persist for long in a wholly quiescent form. Involuntarily—perhaps precisely because of the interest that what is experienced holds—a movement of thought will be released, one that serves to bring out its inner and outer relations. Thus when the poet involuntarily reaches for similes and images in order really to let his subject come to expression. Or when perhaps

it is precisely the contrast with other experiences that serves to illuminate the peculiar character of the single experience. Both likeness and difference can thus serve to cast a clearer light on what was given in the original experience, and reflection therefore does not necessarily stand in opposition to it, but can work in its service and in its spirit, when it has itself vanished.

By the word experience we think chiefly of states in which we were essentially receptive. But reflection can also serve the involuntary life of the spirit, where this appears as urge or striving. In an instinctive or inspired manner, involuntary striving is often led to its goal without any reflection being needed, and perhaps better than could happen with the help of any reflection. When reflection comes to operate in our action, it will be because the involuntary striving met resistance. There may then be needed a deliberation upon what it really is that is being striven for, upon this goal's relation to other possible or actual endeavors, and upon the ways and means that can lead to the goal. Here too it is through the bringing-out of likenesses and differences that reflection renders its service. And here too a difference can appear between dwelling and movement. It may be that the representation of the goal is held fast without great oscillations, and that the relation to ways and means or to other possibilities plays a subordinate role, as when the helmsman stares at the compass or the seamark, and his hand involuntarily moves the rudder according to what he sees. But it may also be that the strongest movement back and forth among various possibilities takes place and must take place before the decisive concentration that is called decision sets in.

The arising of reflection is like a kind of awakening, not from sleep or from unconsciousness, but from the involuntary mental life and its experiences, from states that may provisionally be characterized as sensation, intuiting, representing, mood, emotion, urge, and striving. In its dwelling form, reflection stands closest to the involuntary mental life. In its striving to bring out likenesses and differences, it moves further away from the involuntary. Linguistic usage marks this difference rather aptly by distinguishing between *thinking of* something and *thinking over* something. It is the latter kind of reflection, the discursive, mobile kind, that we shall have most use for in what follows, since it is the one we can best observe, and likewise the one that plays the decisive role in the tasks of thought. Only for this kind of reflection do problems exist. In what follows I therefore understand by thinking this kind of reflection. And yet there is only a difference of degree between the two kinds. Only for a brief moment is a dwelling representing, a holding-fast of a memory-image, possible; precise observation shows that in greater or smaller rhythms we move toward and away from the object we are thinking

of. And every return can bring with it a recognition, so that the relation of likeness comes to play a role just as in discursive reflection. Perhaps the representation was new to us at the first moment; but through repeated dwelling it becomes familiar. Apart from this too, dwelling reflection by no means excludes that manifold other representations and experiences can contribute and perhaps precisely condition the interest that makes continued holding-fast of a single representation possible. What holds for discursive reflection will therefore also hold, to a certain degree, for the dwelling kind.

It emerges from this description that the work of thought consists in comparison, in distinguishing or finding likenesses. When I think over something, I seek to determine its relation to something else, and the more relations of likeness and difference I find, the better I have thought it through. This holds of course also when it is one and the same object that I examine in different connections and under different conditions; I then compare the different ways in which the object appears in the different cases.

But comparison in turn presupposes a holding-together or combining, synthesis. I can compare two objects only when I either have them given to me simultaneously in my sensation, my memory, or my imagination, or else retain a representation of the one while the other is present to sensory observation. If the one object constantly vanished entirely when the other emerged, no comparison could take place. The work of thought consists in a combining by which likenesses and differences become determinative for the relations in which the objects are placed to one another in the connection that results from the work. Synthesis is an expression of energy, of the capacity to perform work.

The resistance that is overcome through this work is due to the objects against which reflection is directed and which are given in the involuntary mental life—in sense-sensation, memory or imagination, feelings of pleasure and displeasure, urge and striving.

The combining work must become the more difficult, the greater the multiplicity of objects that are to be brought into relation with one another. The fullness of content can be so great that the psychic energy available in the particular case cannot master it. There then threatens a dissolution of the mental life, a transition to chaos.

Yet it depends not only on the multiplicity, but also on the degree of difference that the objects present. The more they resemble one another, so that they can fuse, or so that at least the work done in the case of the one need not be repeated in the case of the other, the less work is required. Objects that stand in strong mutual opposition, or perhaps even conflict, require a great energy if they are all to be taken up into the conscious connection of the mental life. This holds both for the inquirer, who must find a point of

view from which the conflicting can stand as one, and for the practically striving person, who must subordinate the conflicting impulses and interests to the thought of a great and guiding purpose.

It will, thirdly, depend on how independent a place the individual objects are to occupy in the connection that is being worked on. A schematic whole is easier to form than a whole that lets the peculiarities of the individual objects each come into their own. Here it is precisely likeness that can create a difficulty, since it easily leads to fusion and thereby to the abolition of the independence of individual objects. On the other hand, from this point of view a pronounced difference or opposition can be a relief.

Furthermore, a greater work is required the further apart the objects lie from one another in time and space. The simultaneous and the near join together involuntarily, and the connection perhaps even forces itself upon us. And if we leave the objects before us in order to immerse ourselves in the world of memory and imagination, here too the connection can often form itself with ease; “with great ease thoughts take up their abode with one another,” but when it is a matter of holding together a present object with distant memories, or with ideals that have been properly to the fore in the soul only in isolated moments of enthusiasm, then special demands are made on psychic energy. To have, in the situation of the moment, the treasures of one’s experiences, one’s reflection, and one’s ideality fully at one’s disposal is perhaps the greatest testimony to mental energy.

Finally, it plays a role how close and intimate a connection the objects, despite their difference, are to enter into. There are here many degrees, from wholly external juxtaposition and grouping to the inner connection that a scientific concept, an artistically formed image, or a great decision can express.

Besides these five conditions—which might be said to form an analogy to what mass is in the case of physical energy—account must also be taken of the speed with which the synthesis proceeds. The genius and the man of will can in an instant expend a capital of energy that other people must distribute over a longer span of time. In the domain of the life of feeling, this opposition appears as an opposition between vehemence and inwardness, emotion and disposition¹, and analogous oppositions hold for the other sides of the mental life. In research, discovery often forms an analogy to emotion, the conduct of proof to disposition.

The psychic energy that expresses itself in synthesis we measure by taking all these circumstances into account. Of exact measurement there is of course no question here,

¹See on this my *Psychologi* IV, 7; VI E, 4–5.

only of an evaluative estimate. But as soon as we seek to give grounds for why one work of thought, one intellectual orientation, one view of life can be called lower or higher than another, we must take all these circumstances into account, and we in fact do so, more or less consciously, and more or less precisely. There is less psychic energy available in the dream-state, where the changing objects each release their own train of representations, than in the waking state, where dominant purposive thoughts bring about a main direction. It is a purely psychological measure that we apply here. And even though, in forming the concept of psychic energy, we have had guidance in the analogy with the concept of physical energy, that concept is nevertheless built upon its own peculiar experiences, in which physical or physiological observations and investigations need play no decisive role. For this reason we need not fail to appreciate the significance of the attempts that have been made, from the experimental-psychological and physiological side, to measure the brain-work presupposed in certain elementary mental activities—e.g. calculating or remembering—through its relation to the muscular work that can be performed simultaneously with it. One can detect such thought-work by the diminution it produces in the muscular work performed at the same time. But this very measurement presupposes that one can, by purely psychological means, distinguish between more difficult and easier mental work; otherwise one moves in a circle. The brain-work corresponding to the mental work described above is, after all, not known directly, but only through the very mental work to which it is assumed to correspond.

3. Reflection presupposes objects on which it can exercise its combining and comparing activity. But now these objects? Does the law of synthesis hold for them as well? We stand here before a question of a peculiar difficulty, closely bound up with the whole nature of the mental life. It is only reflection that acquaints us with the objects. It is the medium through which they are accessible to us. The involuntary mental life in its immediate unfolding must not be described and judged straightway according to the way in which it shows itself to reflection. It is a life within which we are not permitted to presuppose properties and determinations that first come into being through the work of reflection. When, in isolated moments, we are able to immerse ourselves in this involuntary—but not for that reason unconscious—life in such a way that we can afterwards retain a memory of it, it stands before us as a stream, a gliding-along of manifold objects, which we designate as sensations, memories, moods, endeavors, and so on. It is not a chaos of independent objects; already in the involuntary mental life there are connections and wholes as well as differences and oppositions. Every word we use about what we can retain from the involuntary states bears witness to this. Most

striking, no doubt, is the continuity; this is why the word “stream” offers itself so readily for what we call the immediately given. But it is not a uniform stream. Differences of quality make themselves felt; otherwise we could not distinguish between the kinds of objects called above sensations, memories, and so on. And there can be eddies in the stream, or windings, so that the direction is not kept in a straight line. That is to say, strivings of different tendency, of more or less opposite direction, can make themselves felt. With this is bound up, again, that inhibitions can set in, while in other cases, where the different tendencies go in the same direction, a facilitation or an acceleration can appear.

Through the indefinite and figurative expressions that alone stand at our disposal for indicating the basis of objects that reflection presupposes, it shines through that this basis forms no absolute opposition to what emerges as a consequence of the work of reflection. There too we find both unity and multiplicity, both continuity and discontinuity, and we find many degrees and nuances of these relations. It is therefore not justified, with the older English school (*Locke, Hume, Mill*), to conceive the immediately given in the mental life in an atomistic way, as if it consisted in a multiplicity of mutually independent elements. Both in the psychic and in the physical domain, it is only reflection that leads to the formation of atomic concepts, guided by the urge to make the differences as distinct as possible, so that the connection of them can become as clear and perspicuous as possible. The unity to which reflection bears witness, and which it itself seeks to bring about, thereby becomes enigmatic. But it is also unjustified, as in our day especially *William James*² and *Henri Bergson*³ have attempted, to present that involuntary life as standing in the sharpest opposition to reflection and its work. It then becomes, in the last analysis, the indescribable and inconceivable, and it would become an insoluble enigma how there can be any knowledge at all of that “immediately given” from which reflection is nonetheless constantly to draw. James and Bergson represent a significant reaction against intellectualism’s misunderstanding and overvaluation of the significance of reflection, a reaction that in several respects recalls the stand taken by *Hamann, Herder, and Jacobi* against *Kant*. Not pure reason, but the immediate stream of the life of the spirit is, they maintain, the highest, and brings truth forth. The distinctions and definitions of thought are necessary in order that we may practically find our way in life. And analyzing science has carried the opposition to the immediately given still further—an

² *The Stream of Thought*. Mind 1884 (reprinted as Chap. 9 of *Principles of Psychology*). Later especially *A pluralistic universe* (1909), Sect. 7.

³ *Les données immédiates de la conscience* (1888). Later especially in *Introduction à la Métaphysique*. (Revue de Métaphysique et de Morale 1903).

opposition that language itself already favors, by forming independent words for the individual objects and by taking spatial relations as the basis for our intuitive images. It is as though a Fall had taken place when reflection awoke and turned, inquiringly, toward the involuntary life from which it had itself sprung.

Without a kinship between the involuntary mental life and reflection, it would be impossible to understand how the latter could arise. But in the very descriptions that James and Bergson give of what, for them, in sharp opposition to all elaboration by thought, is the immediately given—descriptions that are masterpieces of descriptive psychology—it is nonetheless expressly stated that in this “given” there are differences and connections, qualities and continuities. James emphasizes especially the constant transitions (continuous transitions) and the difficulty of distinguishing between relation and term (relation and matter related); Bergson stresses especially the undivided and yet heterogeneous nature of the inner stream (*durée hétérogène et indistincte; durée dont les moments se pénètrent*). But with this an inner possibility for the arising of reflection has been indicated. The whole weight must not be placed on the external causes (practical need, the mechanics of language, and the technique of science). The motive for the awakening of reflection is given with the opposition between continuity and qualitative multiplicity, between the connection and the individual elements. Herein lies the possibility of an inhibition of the involuntary stream, an inhibition that only discursive reflection can remedy, through the clarity and definiteness it is able to bring. As a combining that is accomplished through comparison, reflection can bring about clarity concerning the relation between continuity and discontinuity, unity and multiplicity. And it is not anything wholly foreign that it brings in by its work; for already in the immediately given there appear, after all, likenesses and differences, connection and multiplicity—and likewise, what is especially significant, an involuntary striving to bring about harmony among the different tendencies. If there were not such an original striving, it would be unjustified to speak of a stream, of a *continuité mouvante*, a *unité de direction*⁴! There shows itself here, then, an analogy to what, in the case of reflection, we called synthesis—a tendency to gather and unite. And reflection awakens when there is a special urge for a fuller carrying-through of this tendency than is possible within the involuntary mental life. Already within this—not only after reflection has

⁴One of my young listeners wrote to me some years ago: “I can declare myself free of every purpose, as of every interest or belief in anything outside or inside me. I do indeed feel something willing, something pressing forward, but it lives, as it were, its own life, pushes aside everything that gets in its way, and surely never commits any act that is not to its own advantage.”—Here reflection has brought no result, only dissolution; but the involuntary life has maintained itself in its unity and in its definite direction, which was all the more striking to the young psychologist the less his reflection could attain to any firm standpoint.

awakened—crises and catastrophes can arise, and chaos can threaten. A work can be required even before reflection can awaken; but the difficulties arise, are worked on, and are resolved on different terms in the involuntary mental life than where reflection has taken the helm. Perhaps some of the most significant things in our mental life take place in the regions below the domain of reflection, and it is often only the after-effects of these deep-lying processes that reflection has to deal with. It will always be impossible to demonstrate that at any point we stand before an immediately given in the absolute sense, that is, one in which absolutely no psychic work has been done. What we in each particular case call immediately given deserves this name only in relation to a certain definite psychic work, while in comparison with other objects it would perhaps have to be regarded as the result of a work.

4. By starting from reflection and seeking to become clear about its way of operating, we arrived at the concept of synthesis, and we then found that this concept could also be applied to the involuntary mental life, which forms the constant basis from which reflection springs, in order thereupon to make that basis itself an object of its activity. We must make ourselves familiar with the fact that the involuntary mental life, as it unfolds prior to the awakening of reflection, can be illuminated and designated only by way of analogy. We apprehend it only in analogy with clear reflection, and the expressions of language are throughout drawn from the world of reflection. So far is that involuntary mental life from being the thing easiest to apprehend, that it takes precisely great art to get a proper view of it in its difference from the mental life shaped by reflection. What one calls sound common sense operates, in its psychology, with concepts drawn from the world of reflection. A childlike tendency makes itself felt to regard everything in the mental life as the fruit of clearly conscious thinking, of deliberation and calculation. One personifies, so to speak, not only outer but also inner nature, placing a person behind the person. The childlike psychology explains personal life itself, in its involuntary unfolding, by supposing that behind it there is a person who consciously thinks whatever stirs in it—just as in older works on optics one can find depicted a man inside the head, looking at the image that the light-rays form on the retina!

Reflection is required in order to discover and apprehend the involuntary mental life, and yet the peculiarity of the latter consists in the fact that reflection plays no role in it. This difficulty can be resolved only by our apprehending and expressing the involuntary—insofar as any memory of it persists after the awakening of reflection—in analogy with what we know clearly from the world of reflection itself. This is what we do when we say that synthesis is the fundamental form, not only for comparing

reflection, but also for the involuntary mental life, and is thus the most comprehensive psychological concept.—

But now a new difficulty arises. Synthesis means combining—but there must then be something that is combined, and this something must be a multiplicity. The objects toward which reflection directs itself in order to hold them together and compare them form a coherent multiplicity; each individual object comes into being for us through an act of distinguishing. And in a single object we can again distinguish individual sides, properties, or parts, which we may call elements. The result seems then to be that we end in an opposition between synthesis and elements, and that the multiplicity of elements is the constant presupposition for synthesis being able to operate! From this one might then draw the conclusion that what is original in the mental life was a multiplicity of elements, and that only at a later point of development did the combining activity intervene. This conception was also asserted in its day by John Locke, and it still often haunts psychology. It relies on the analogy with the formation of a mechanical whole out of elements that could be pointed out beforehand each by itself, as when a heap of sand is formed from grains of sand that previously lay each in its own place and are only now brought together. The question becomes whether this analogy is justified.

When reflection awakens, the mental life does not, however, begin with it; for then it would have no objects. The elements—that is, the parts, sides, or properties that reflection can distinguish in the object or objects, and that it can determine through their mutual likenesses and differences—are, after all, precisely the fruit of a work of thought. They are drawn out of the connection in which they involuntarily occurred, and set in a certain opposition to one another—an opposition without which they could not become objects of clear consciousness. And the characterization that each individual element, and thereby each individual object, receives is conditioned by its relation to other elements or objects. At no point is it justified to assume isolated elements in the mental life, each absolutely independent by itself—a kind of absolute psychic atoms. That is to confuse the effect of reflection with reflection's own basis—at bottom, again an example of the naive psychology that tends, after all, to confuse differences that thought first makes with originally given differences.

And insofar as we succeed in retaining a memory of the state we are in when involuntariness prevails, this state appears—and here the masters of descriptive psychology are undoubtedly right—not as a chaos of independent parts, but as a stream, a pulsation, yet not without differences, but such that there is a connection even between the lowest troughs and the highest crests of the waves. We find, as far as can be judged, unity and

multiplicity so interwoven with each other that they cannot be separated in an external manner. It was, after all, precisely in the urge to gain greater clarity about their relation that we found the possibility and the motive for the awakening of reflection.

At no element can we, in the domain of the mental life (any more than in that of matter), come to a stop as absolutely isolated or absolutely simple. There constantly poses itself, for reflection, the task of finding a connection between the elements or objects that might seem to be isolated, and of finding an inner multiplicity in the elements or objects that appear as simple. It is a sign of the dullness of reflection when it comes to a stop too early in either of these respects. In other words: the law of synthesis must always be carried through anew with respect to the objects or elements themselves.

On the other hand, neither can there be any absolutely conclusive synthesis. As long as the mental life lasts, new objects are taken up into the involuntary stream and are involuntarily worked into it. They are new drops, or perhaps new brooks, which are partly reshaped by the stream and partly can also gain influence on its character and its course. To this is added the character of the new riverbeds that the stream must form for itself when the terrain changes. And as long as reflection is awake, it will constantly be able to set itself the task of bringing about a greater, more comprehensive, and more intimate synthesis than before. There is always a psychic work to be done. The synthesis that has hitherto enclosed the multiplicity of the mental life can again come to stand as object or element for a later synthesis. As soon as we become conscious of our preceding mental states and activities, such a transition to a new synthesis takes place, whether or not this signifies a progress, a more valuable state. It is a new I that becomes conscious of our earlier I's. No synthesis is definitive. Even the philosophy that spoke most of higher unity always fell into embarrassment when it had to speak of the highest unity.

It can of course be entirely justified, in a special connection, with a view to a particular investigation, to disregard these difficulties of principle and to regard elements or objects as if they were absolute and simple. Every special scientific investigation presupposes a certain abstraction. If, e.g., experimental psychology really needs to operate with a kind of psychic atoms or flashes as independent and simple, this can be just as justified as when, in natural science, one operates with material atoms. But one must avoid the dogmatism that confuses concepts we have arranged for the needs of an investigation with ultimate realities. And likewise it can be justified that in everyday speech, or in a practical—e.g. ethical or juridical—respect, we regard the individual as an absolute unity with a completed synthesis, without regard to the fact that synthesis at bottom signifies a work, an activity, not something given once and for all. There is an irrational

relation between the mental life and what we call its elements. The point now is to find as many elements as possible; but it can be justified and necessary to confine oneself to a few decimal places, provided only that one does not forget that the question is then not exhausted. The irrational is here, as everywhere, both a barrier and a task.

There expresses itself here an antinomy that is closely bound up with the essence of the mental life and is peculiar to the concept of personality. Mental life and personality cannot be conceived as products of previously given elements, since the elements we know always have their properties precisely by being members of the mental life. We here come to move in a circle, unless we are clear that an inexhaustible reciprocal action prevails: the parts are determined by the whole, and the whole by the parts. In each particular case, in each particular psychological consideration, we must investigate how far the synthesis, the formation of the whole, has advanced, and on this basis observe how the emergence of a new object operates. But the very distinction between synthesis and object is always more or less artificial in relation to the constant interplay of meeting and parting, of continuity and discontinuity, of wholeness and dissolution. There is here a mental rhythmicity that forms an analogy to—or rather a prototype for—the rhythms that characterize the course of organic life, a rhythmicity that by no means begins with reflection, but, by all we can judge, already makes itself felt in the involuntary mental life. Indeed, this rhythmicity recurs in the very relation between reflection and the involuntary course of the mental life. The difficulty of understanding the origin of the mental life rests precisely on the fact that it cannot be explained as a product of previously given elements, but from its very first moment moves in a rhythm—and that in a rhythm of the peculiar kind that none of the members in the rhythm can have absolute priority over the other.

5. In the history of the concept of synthesis this antinomy has been a stumbling-block. The concept of synthesis itself is an old idea, traceable in Plato and Aristotle, and especially in Plotinus in antiquity, and which from Leibniz's time on has always, with more or less energy, been asserted in psychology and in philosophy generally⁵.

Hume cut the knot by dissolving the mental life into absolutely independent elements, and consistently he therefore found an insoluble enigma in all unity and connection in consciousness. The principle of unity (the uniting principle) was for him the unintelligible. *Kant* started from the opposite point of view, that it is precisely the unconnected, the

⁵On the history of the concept of synthesis in more recent times, see *Georges Dwelshauvers: La synthèse mentale* (Paris 1908), pp. 179–223 (an account of the concept of synthesis in Leibniz, Kant, Wundt, Høffding, and Pierre Janet).—In *S. Kierkegaard* this concept appears in an energetic and profound manner, especially in the measure he applies in his doctrine of the stages. See my book *Danske Filosofer*, pp. 153 f.; p. 165 f.

isolated and mutually independent, that is unintelligible to us, both in psychology and wherever else we meet it. But he still held fast to Locke's presupposition that, as the basis for thought's combining and ordering activity, there was given a multiplicity that was only now to be connected. Thereby he arrived at his fateful distinction between the form of cognition and its matter. This too was to cut the knot. And yet Kant did not lack the insight that already in the involuntary mental life there operate forms analogous to those that clear reflection employs. Synthesis is, he says, the effect of a blind but indispensable function of the soul, without which we would have no cognition at all, but of which we only rarely become conscious. It is the task of thought to form the synthesis into a concept. And this same blind function of the soul (Kant calls it imagination) is likewise what conditions the connection between the two extremes of our cognition, "sensation" and "understanding." It was surely only Kant's reluctance to give psychological considerations too large a place in his investigation that held him back from giving a fuller characterization of the involuntary course of the mental life. J. F. Fries developed Kant's doctrine further from this side and is even regarded by his disciples as the real discoverer of the involuntary basis⁶. But he then believed that in this basis—the spontaneous striving, the immediate cognition—he possessed a rule of truth exalted above all error. Unfortunately this rule is ineffable, since it can come to consciousness only in an imperfect and fragmentary way through reflection! A dogmatism comes to the fore in the firm belief in the validity of the involuntary mental life—a dogmatism that signifies a decided step backward in relation to Kant. Already when Fries assumes that "spontaneity" is not only continuous but also constant, he says more than any reflection could demonstrate. The irrational relation between immediacy and reflection is not strictly maintained.

This relation is, as already mentioned, very sharply emphasized by *Henri Bergson*. This thinker is not satisfied with the concept of synthesis, because it always presupposes an opposition between unity and multiplicity. Is it not an opposition—he asks—that recurs, whatever one thinks of or speaks about? What matters is to say *which* kind of unity and *which* kind of multiplicity we have in the mental life. The streaming life of the personality cannot be defined by means of concepts that always stand in a certain relation of opposition to one another. One cannot, through thesis and antithesis, arrive at a real unity. One must, conversely, immerse oneself in the immediate stream of the mental life in order to see the opposition emerge from it. Immediate beholding (intuition)

⁶Against this, see my review in *Göttinger Gelehrte Anzeiger* (1907) of "Abhandlungen der Frieseschen Schule. Neue Folge."

must take the place of reflection⁷.

Bergson's standpoint recalls, viewed from one side, that of Fries, except that he gives us, with far greater mastery, an understanding of what is meant by "spontaneity." But whereas Fries and his disciples held that true cognition is attained by translating, as well as we can, that spontaneous basis into the forms of reflection, Bergson demands that we subdue reflection and, by an act of will, dive down again into the immediate, feeling ourselves one with it. It stands here as an enigma how the validity of a cognition can be demonstrated without reflection. One of the special problems of thought will later give us occasion to bring out this difficult point in the philosophy of the brilliant French thinker.

In response to the remark that the concepts we use in our reflection to characterize the mental life—synthesis, unity, multiplicity, and the like—find application to all objects, I will maintain that the reason why that group of concepts constantly finds, and must find, application to all objects is to be sought precisely in the fact that they express the forms in which the mental life, and cognition in particular, unfolds itself. Only through the application of these forms does the appropriation, the taking-up into the connection of our own being, in which all understanding, all cognition consists, become possible. No wonder that all products of cognition bear their stamp! Our cognition consists in a conscious or unconscious analogy with our own being.—

The relation of opposition that has been discussed in the foregoing under various expressions (object and reflection, matter and form, spontaneity and reflection, intuition and analysis), *Stumpf* designates as a relation of opposition between phenomena and psychic functions⁸. With respect to this relation he advances, with all possible reservation, the hypothesis that psychic phenomena are not altered because they are analyzed. When we become aware of something in an observation that we did not notice at the first moment, this is supposed to bring about no change in our observation itself, but only in the degree of explicit consciousness we apply to it. A sound rings the same whether or not I notice individual tones in it; a dish makes the same impression whether I can distinguish something sour or sweet in its taste, or whether the taste acts as a unified whole. In my judgment it is, from the outset, not probable that this is how matters stand. The differentiated and differentiating attention in which analysis consists must alter the object for us. The fact that we have discovered a spot in an otherwise beautiful picture—a spot we did not notice at the first moment—can give the whole picture a

⁷*Introduction à la Métaphysique*. (Revue de Métaphysique et de Morale 1903, pp. 15–17.)

⁸*Erscheinungen und psychische Funktionen*. (Abhdl. der preussischen Gesellschaft der Wiss. 1907), pp. 15–21.

different character. The timbre changes; an element has, quite simply, been added that was not present in the first glance. And the relation between the individual parts is in each case a different one after the analysis than before. It can cost an exertion of will to return to the immediate beholding in which the particulars have not yet become the object of special apprehension. There is here a difference between experience and reflection that can scarcely be effaced. There are of course many degrees, as we shall come to see in what follows, in the investigation of the relation between intuiting and judging. And individual differences certainly also make themselves felt. In elastic natures the transition from experience to reflection, and vice versa, will take place relatively easily; and yet I should think that even in them a certain contrast-effect will make itself felt between the two states.

Stumpf, for that matter, will only maintain the possibility of a differentiated subsistence, during the experience, of what is first differentiated with consciousness during reflection. And he concedes that there is probably no absolute independence in the relation between phenomenon and function “under the usual, complicated circumstances of the psychic process.” Whether experimental investigations can, as he surmises, lead to any result here is doubtful. It is not easy to see how the difference between immediate experience and reflection could fall away. And in a psychological experiment it is difficult to obtain a wholly immediate experience.

Chemists often say that the atoms of the elements are present in compound substances—so that, e.g., carbon and oxygen are really present in carbonic acid. And *Stumpf* holds that the psychologist must have the same right in his domain. I think, however, that there is a difference. For the chemist can demonstrate that there are just as many parts by weight in carbonic acid as in carbon and oxygen each by itself; and at bottom that is all he means by that manner of speaking. Such an equivalence the psychologist is, for good reasons, debarred from demonstrating.—For that matter, a chemist such as *Sir William Ramsay* has expressly denied the right to take that manner of speaking literally, and *Ernst Mach* has expressed himself in the same direction. The analogy is therefore doubtful.—

We do not get beyond an antinomy between experience and reflection, and we may not, without further ado, transfer to the one what holds for the other. Experience itself we may think of neither as absolute continuity nor as a chaos of differences. The continuous and the discontinuous occur in it in all possible degrees, and it is only reflection that takes the opposition between continuity and discontinuity sharply into view, in that it—as will later be shown—receives the double task of finding differences where continuity was

immediately given, and intermediate links where discontinuity was immediately given.

6. In the concept of synthesis there lies, as we have seen, an opposition between object and form, which gives rise to great difficulties that become insuperable when by object one understands an absolutely unformed matter. As already indicated, every object will pose a double task for combining thought: to determine its relation to other objects, and to find out differences and relations within the object itself. Hereby arises the constant striving to make the discontinuous continuous and the continuous discontinuous.

But when we do not separate thought from the other sides of the mental life, there is yet a third element, besides form and object, that we must bring out. When a task is posed, there is presupposed an urge to have the task solved. The work on the solution then acquires value, since we designate as valuable everything that satisfies an urge, whether this urge expressly announces itself, or whether it is only from the satisfaction we feel at the work or its result that we infer that there has been a lack that has now been filled.

Thought has biological significance, in that attentiveness to changes in the surrounding world makes it possible for dangers to be averted and advantages gained. But here too the significance of synthesis shows itself, since it is only under very elementary conditions that attentiveness to a single change is sufficient; it will be of importance that the single experience be brought into connection and reciprocal action with other, present or earlier experiences; only then can the reaction to what is experienced become entirely appropriate. Only instinct reacts to an elementary stimulus; when reflection is awakened, a more serious orientation, and thereby a richer relation to surrounding existence, is possible. An interesting example of a synthesis that is perhaps of a purely physiological kind is afforded by what *Pavlov*⁹ has called conditioned reflexes. Salivation can be produced not only by direct stimulation with a substance brought into the mouth, but also by stimulation of sight, hearing, smell, or the skin-sense, on the condition that these sense-stimulations have not occurred too long beforehand together with the action of that substance. Here an associative synthesis is at work, which makes it possible for one kind of stimulation to take the place of another with which it originally had nothing to do. Already in so simple a case as this there takes place a combination of present and earlier states. What shows itself in this simple example repeats itself at the higher stages of psychic development.

Immediate value belongs from the first to the result—in the given example, salivation (which of course, as we may here disregard, itself again has its value as a member of the

⁹*The Lancet*, 6 Oct. 1906.

digestive process). The sense-sensations that indirectly produce it acquire at first only mediate value. And at more developed stages it will, similarly, be the result of reflection that has value, while reflection itself is only a means. But it is quite possible that a value can pass over from being mediate to becoming immediate. There then occurs what we in psychology call a displacement of motive.

When it is bitter need that leads to thinking, the value or motive stands in a purely external relation to thought. The value is here itself an object, since, as it cannot be immediately attained, it poses the task of finding ways and means, and thereby comes to stand as a purpose. In the definite and compelled relation between purpose and means we have man's first encounter with the necessity of thought. If he wills A (the purpose), he must also will B (the means). By this path man enters the world of reflection even where there are no more direct motives for entering it. But it can acquire an independent value, can become an object of direct interest, to investigate that necessary relation between purpose and means, which is, after all, the same as the relation between effect and cause; and the value is then bound to reflection itself and its work. It comes to stand as an activity that it is attractive to exercise, whatever result it may lead to.

When a stage has been reached where the value or motive attaches itself closely to the activity of thought itself, so that this is both end and means—when, in other words, an intellectual feeling has developed, a joy in thought itself, as it unfolds according to its own laws and according to the demands of the objects—then we need not specially recall motives and values in the case of cognition. We then desire no other satisfaction than that which is bound to the use of our faculty of thought.

It is such a stage that we presuppose in an investigation of human thought. All purposes other than that of seeing where consistent thinking leads then do not exist, and we need not constantly recall the standpoint of value; this is given once and for all together with the very urge to—observe and to infer. Even where other motives than the intellectual interest itself actually make themselves felt in a movement of thought, it will nevertheless be fortunate if such motives—in relation to which the work of thought stands only as a means—pass over, through displacement of motive, for a while into purely intellectual motives. Thereby the work is best done. And it will in any case always be, in the end, an examination undertaken from purely intellectual motives that must revise the work performed from other motives, whatever these may have been.

7. The concept of energy in the sense in which we have here taken it has its origin in natural science, and it is by way of analogy that we have attempted to transfer it to the domain of the mental life. All psychological expressions are, after all, transferred from

the material to the spiritual, from the world of space to the world of spirit and thought. But there are difficulties in carrying the analogy through. An exact measurement is possible in the psychic domain only as regards relatively elementary mental phenomena, and never in so perfect a manner as in the case of material energy. Modern physiology stands, as regards method, on the standpoint that it requires physiological states to be explained by physiological causes, insofar as physical causes from the surrounding world do not intervene. Every state of the brain or brain-activity is therefore also to be explained on the basis of the earlier states of the brain and of the organism. According to strict natural-scientific method, every material state within and outside the organism must find its explanation in being seen as having arisen through the conversion into new form of the physical energy that expressed itself in preceding material states. When two brain-states thus succeed one another, the explanation will therefore not be found until it is demonstrated that all the energy that expressed itself in the first state has, without remainder, been converted into the second state, insofar as it has not radiated out into other physiological processes. It is perhaps impossible to carry out experiments in this direction; but such experiments would mark the completion of the physiological method, which can set as its motto *Spinoza's* words, that an explanation of a bodily state by the intervention of the soul is one with an admission that one cannot explain it. Even if now one of the experiments one might make seemed to show that, in the arising of an organic state, physical energy arose or vanished without a physical equivalent, one would, from a purely natural-scientific standpoint, sooner believe that there was an error in the experiment than rest content with having discovered a limit to the conservation of physical energy. "The principle of the conservation of energy," says *Maxwell*¹⁰, "has acquired so great a scientific weight that no physiologist would place any confidence in an experiment that showed a considerable difference between the work performed by a living being and the sum of the energy received and expended." Since Maxwell wrote down this proposition, experiments have been carried out which—within certain limits of error—show that the exchange of matter and force during the performance of mental and bodily work presents an equivalence between what the organism took in and what it expended¹¹. It will hereafter become even more difficult than before to assume that in any part of the organism something should take place that conflicts with the energy principle.

But how, under these conditions, will it be possible to maintain psychic energy as

¹⁰*Scientific Papers*. II, p. 759.

¹¹*Atwater: Neue Versuche über Stoff- und Kraftwechsel des menschlichen Körpers* (Ergebnisse der Physiologie III, 1. 1904).

a peculiar form of energy? Is it not, then, more reasonable to assume that all energy is physical, and that the psychic energy we have spoken of in the foregoing is only a symptom of a physical energy to which we have no direct access?—For we cannot carry through the principle of equivalence within the psychic domain itself, as we can in the physical domain. Interruptions occur within one and the same individual conscious life (through sleep, faint, and so on), and between different individual consciousnesses we cannot demonstrate psychic continuity, as we can demonstrate physical continuity between different bodies—e.g. between two organisms.

Accordingly, the demand has been raised that the psychological definitions should be replaced by physiological ones. Among us, *Carl Lange* (in his “Nydelseernes Fysiologi”) has asserted this demand. In strict epistemological and psychological form, *Richard Avenarius*, in his acute work *Kritik der reinen Erfahrung*, has maintained a similar view. Full scientific understanding is attained, he holds, only when the psychic states are traced back to the corresponding physiological states: hence he calls the series of psychic states the dependent vital series, the series of physiological states the independent vital series. And his justification is that only in this way can we reduce the qualitative differences that the psychic phenomena present to quantitative ones, so that an exact formulation of the relation between the changing states becomes possible. The completion of our cognition will thus consist in a reduction of the psychic to the physical¹².

In the face of the views that thus demand a reduction of psychology to physiology in order for a scientific psychology to become possible—views that thus would really abolish psychology in order to make it a science—it must be acknowledged that the actual discontinuity and the qualitative difference among the psychic phenomena will always set limits to a strict carrying-through of psychology as a science. But the question is whether the path they point out leads to the goal—whether the method they praise does not at bottom contain a palpable self-contradiction.

When it is demanded that the psychological definitions should be replaced by physiological definitions, the presupposition for this must surely be that the psychological definitions are complete and exact. But to provide these definitions is psychology's own affair, and if it cannot provide such definitions, physiology cannot know what it is that it should seek to explain in the brain. When that which is to be replaced is vague and wavering, the replacement must turn out accordingly. There will be no certainty that one has really found the brain-states corresponding to the definite psychic phenomena one has in view. The independence of psychology must therefore in any case be

¹²Cf. my characterization of *Avenarius*'s philosophy in *Moderne Filosofen*, pp. 98–106.

acknowledged, since it is to give us the symptomatology that sets physiology its tasks. Now, for every empirical science it is a long and difficult work to arrive at conclusive definitions; such are possible only when the empirical science is essentially finished; they come at the end, not at the beginning, of the investigations. All too often one operates with unfinished psychological definitions that are regarded as reliable starting-points for brain-physiological interpretation. Thus *Descartes*, and in more recent times *Lotze*, regarded the very concept “soul” as so clear and fixed that one could confidently call upon physiologists and anatomists to search for “the seat of the soul.” And in the most recent times *Flehsig*¹³ has regarded the concept of association of representations as so simple and clear that he could suppose he had found special tracts in the brain for processes that are supposed to bring about the connection between the physiological correlates of the individual representations. It is in reality a great abstraction that Flehsig works with when he sets the connection itself (the association) in opposition to the elements (representations) that are to be connected, so that the former could be thought of as localized in places other than the latter. He thereby quite overlooks the psychological antinomy brought out above, which warns against confusing without further ado the differences and distinctions that the investigation necessitates with oppositions in the unfolding of the mental life itself. There is no “element” in a mental life that does not stand in, and is not determined by, some connection or other, some relation or other. When a representation resists entering into a certain, seemingly natural association, the reason is to be sought in the fact that it is a member of another, more intimate association and cannot be detached from it. The conditions here are very complicated, and with a purely mechanical schema neither physiology nor psychology comes into its own.

But let us suppose that the psychological definitions are complete and perfect. Even then great, unsolved problems will remain. How can physiological states have psychological symptoms? How can qualitative differences correspond to quantitative ones, and how can discontinuous phenomena be bound to continuous processes? Far from such questions falling away through the “reduction” of psychology to physiology, they are precisely sharpened by it and appear in a more glaring form. And in any case it stands as an unsolved enigma how the qualitative differences and the discontinuities come into being. Even if one should succeed in showing that two phenomena, A and B, which one has hitherto regarded as different, are in reality identical, so that one can set $A = B$, one has not thereby explained how A and B could in the first place stand as different. To call this difference “subjective” does not help; it is not thereby gotten rid of out of existence. It

¹³*Gehirn und Seele*. Leipzig 1896, p. 24 f.

is, after all, precisely the fact that there is something “subjective” in the world that makes psychology both possible and necessary. It is, moreover, within this “subjective” that all cognition of material nature—physiology too, and with it the knowledge of our own body, especially of our own brain—comes into being. Cognition is itself a psychic function, both where it appears as sensation and where it appears as thinking. By regarding psychic states and activities as accidental appendages (“epiphenomena”), the natural scientist would saw off the branch on which he sits. To study this branch itself is, to be sure, not his task; but he does well, nonetheless, not to forget where he is sitting.

When *Avenarius* attempts to carry through the reduction of the doctrine of “the dependent vital series” (psychology) to the doctrine of “the independent vital series” (physiology), it is nonetheless apparent through his exposition that he constantly infers from the “dependent” to the “independent” series, not the reverse. It is on the basis of the psychological observations he has collected that he constructs the physiological schema which for him is the proper object of science, and one cannot understand what is said about the “independent” series except by means of the facts that the “dependent” series contains. In general it turns out that brain-physiology is still far from developed enough to give a thorough elucidation of the material processes to which the great, richly nuanced world of psychic processes is in one way or another bound. What one finds in this respect among physiological psychologists and experimental psychologists are mere schemata, which are supposed to “explain” what psychological experience presents. No one has more occasion to experience this than a psychiatrist who is at the same time an experienced and critical psychologist. Thus *Pierre Janet* has declared that, apart from individual observations and histological investigations, the physiological interpretations are only translations of the psychological concepts into another language, and that if one will think only anatomically, one must wholly give up thinking psychiatrically¹⁴.

8. It is the discontinuity and the qualitative differences that cause psychology’s difficulties. But greater weight is often placed on them than experience warrants. Precise reflection often leads to the discovery that even where there seemed to be a psychically empty space, there was in reality a psychic content given, although it was not grasped by attention or by recognition, or was forgotten immediately after the experience. And as regards the qualitative differences, observation will often find more and finer nuances than those one first stopped at and perhaps confidently declared to be utterly disparate. The continuity in conscious life is surely greater than is often assumed, and there can lie a scientific danger in too strongly proclaiming discontinuity and qualitative difference

¹⁴*Les Obsessions et la Psychasthénie*. I, p. 496; 605. Similar statements by *Wundt* (*Philos. Studien* XIII, p. 363) and by *Ramon y Cajal* (cited in *Herrlin: Minnet*, p. 21).

as the distinctive mark of the phenomena of consciousness. The inner connection in consciousness—the fact that the more we immerse ourselves in, and make ourselves one with, the involuntary course of the mental life, the more this appears to us as a whole, a streaming—is and remains the primal phenomenon in the psychological domain. We overlook it so easily only because in everyday speech, and on the whole in practical life, we have most use for certain single drops or waves in the gliding stream. We must perform an integration when we go back from our more or less mechanical practice to the immediately given. And even if we confine ourselves to functions that can be exercised with clear consciousness—e.g. memory and comparison—these very functions bear witness to an inner connection behind the particulars. They are forms of synthesis, of the tendency to combining and holding-together that attests the mental life's constant striving to be a whole. The unconnected is the psychologically enigmatic, that which sets psychology its problems. The task of psychology is to find connection between the elements of consciousness as well as between its different states.

Leibniz has the merit of having energetically maintained the principle of continuity in the psychic as well as in the physical domain, especially by pointing to the elements that are vanishingly small in relation to the clearly conscious states, and that only by being summed up and combined produce a result accessible to untrained self-observation. Already a comparison between modern and ancient or medieval poetry shows how many more mental nuances self-observation has now come to perceive. It is scarcely only because the nuancing in the psychic domain is now in itself greater than in earlier times that this greater richness is found. It is partly precisely the knowledge and understanding of the immediate and involuntary in the mental life itself that has grown. The childlike and primitive psychology is inclined to conceive everything that takes place in the soul as clearly conscious and all actions as undertaken with deliberation and intent. Now an explicit demonstration is required that reflection and deliberation have been present in the particular case. The transitions themselves between the unconscious and the conscious, and between the involuntary and the voluntary, can take place through many intermediate forms, many degrees. The more practiced self-observation and criticism are, the more difficult it is found to be to put one's finger on a definite place where the boundary should lie.

Leibniz was inclined to regard all psychic differences as differences of degree with respect to clarity and obscurity. This attempt was characteristic of the century of the Enlightenment, which Leibniz inaugurated. But there are other paths by which psychology can find a continuous connection despite the qualitative differences. There

constantly take place fusions and compositions in the domain of conscious life—of sensations, representations, feelings, and endeavors. And displacements take place by which what at first stood in the foreground is supplanted by what was at first background—where what at first had only mediate value comes to stand as immediately valuable. And in integral characters, the highest and noblest objects of psychology, there appears to contemplation so intimate a dependence of everything that stirs in the soul on one purpose, one guiding thought, that we have here a causal connection that is every bit as firm and definite as any in the physical domain. Every single representation, every single mood, and every single impulse is, in such a character, clearly determined by its place within the whole to which it belongs and which it itself helps to shape.

Where, despite everything, a lack of connection shows itself in the psychic domain, we need not, however, give up the principle of continuity, any more than the physicist gives it up when energy seems to have vanished in a state of rest or equilibrium. Just as the physicist introduces the concept of potential physical energy to express the fact that what seemed to have vanished can be released again, so the psychologist will be justified in speaking of potential psychic energy, supporting himself especially on the fact that psychic elements can be reproduced after an interval in which they have not been in consciousness. Words such as “predisposition,” “possibility,” “disposition,” “capacity,” “trace,” “tendency” express the same as the word potential energy. The psychologist is, however, worse placed in applying this concept than the physicist in applying his corresponding concept. In the first place, the physicist can directly point to the phenomenon in which the potential energy has its seat—e.g. the tensed spring or the stone resting on the roof. In the psychic domain, by contrast, complete equilibrium (rest) means the same as unconsciousness—hence something inaccessible to observation (whatever it might otherwise be in itself). And in the second place, in the physical domain one can carry out conversions that show that a quantum of energy can be released corresponding to the quantum that apparently vanished on the transition to the “potential” state. But in the psychic domain it cannot be demonstrated in this way that predisposition and development correspond to each other. If the concept of potential psychic energy nevertheless has its justification, it is because it stands as the expression of an unsolved but undeniable task, and because it contains a summons to go ever further in observation and analysis in order to find as much continuity as possible between the different states of the mental life. There is a long way to absolute rest—and it might be that it occurs as little in the psychic as in the physical domain. Possibility everywhere subsists in an actuality, whether or not this is accessible to us.

9. The only way in which psychology and physiology can work together without either of them coming to break with its guiding thoughts is provided by the psychic phenomena being regarded as symptoms of physiological states—and, conversely, physiological phenomena being regarded as symptoms of psychic states. We can then *infer* from the one series of phenomena to the other, although we cannot *derive* the one from the other.

It is the natural-scientific method that step by step leads to such a conception. The law of inertia requires that every change in the state of a material point (of rest or of rectilinear motion with a certain velocity) be derived from the influence of another material point. Thereby it is excluded that a non-material (that is, non-spatial) cause can acquire natural-scientific significance. Thereby the principle is established that a material phenomenon must have a material cause. *Maxwell* finds the experiential proof of the law of inertia in the fact that, every time we encounter a change in a body's state of motion, we can trace this change back to an action between this body and another body. This experiential proof natural science is constantly in the process of furnishing.—To the law of inertia is joined the proposition of the conservation of physical energy, which, as already mentioned, seems to have to be maintained also in the case of organic nature.

There are in the physics of our time intimations that the said propositions, despite their great importance for the whole conception of nature, are nevertheless derived from laws and relations that hold for the ether-vibrations in which one sees more and more the fundamental thing. Philosophically regarded, this would change nothing in the conception of the problem of the relation between the spiritual and the material. The vibrations of the imponderable ether are just as different from the psychic phenomena as the movements of the ponderable atoms. The problem remains the same, even if electrons take the place of atoms. It is the spatial character that, for philosophy, is the decisive property of matter; in this *Descartes* saw rightly. It must not here mislead us that physicists sometimes use the expression the immaterial in the sense of the imponderable.—

If now this is so, there can be found no place for the mental life in the series of physical, or respectively physiological, phenomena. The only possibility then becomes the conception that in the mental life we have the expression that self-observation gives us for what appears to our senses as organic processes. Soul and body are then not two different things or beings, but two languages in which existence speaks to us. We can perhaps translate from the one language into the other; but we cannot derive the one language from the other.

Since the mental life cannot be derived from the laws and forces of material nature, we must assume that in all material activity there cooperate, besides the properties whose presence natural science demonstrates, other properties that are not accessible to sensation, and in which the mental life would be able to find its explanation. Philosophically regarded, natural science's concept of matter cannot exhaust existence. But by the "predispositions" or "dispositions" that we presuppose in order to understand the possibility of the mental life's arising in nature, we do not in the least shake the application that natural science makes of what it has learned and seen.—

The conception to which we here adhere really returns to the childlike and practical view that regards the body as the outer expression of personality, accessible to the senses. And we thereby obtain at the same time the best and most durable working hypothesis, in that we are summoned to follow at once the series of psychological and of physiological phenomena, each by itself, as far as is at all possible, and to investigate the relation between them, the analogies and proportionalities, as exactly as possible. It is a functional relation, in the mathematical sense of the word, that this conception assumes between the mental and the material phenomena; we seek, and often we also find, that a variation of the mental states corresponds to a variation of the bodily states, especially the brain-states, and vice versa. And to this is in reality confined the knowledge we have of these things. Our hypothesis (for which the most fitting name is the identity hypothesis, since it does not assume that soul and body are two different beings) we regard provisionally only as a working hypothesis. Whether it gives a conclusive explanation of the relation between the spiritual and the material, between psychic and physical energy, is another question, which will be discussed in a later connection. Even thinkers who reject our hypothesis as a definitive solution nevertheless concede that it is a useful working hypothesis. Thus *H. Bergson* and *C. Stumpf*. Whether this concession is consistent from their standpoint is a question in itself. A hypothesis that has no ground at all in the nature of things will not in the long run be suited to serve as a working hypothesis.—

It has sometimes been thought that the enigma could be solved by assuming (or saying) that psychic energy is nothing other than nerve-energy. But when one does not assume that all nerve-energy is connected with psychic phenomena, a distinction must be made between conscious and unconscious nerve-energy—and the problem is then posed anew. Besides, we know as good as nothing about this so-called nerve-energy and about its relation to other forms of physical energy, and it is of no use to give the nerve-energy to which psychic phenomena are bound a special name. Thereby one only introduces a *qualitas occulta* and rests content with it. The natural-scientific concept of

energy is, wherever one meets it, always formed on the basis of phenomena with spatial, geometrical properties. Natural science knows energy only as the expression of relations between phenomena in space. And we cannot, unfortunately, form a concept of energy from which the physical and the psychic concepts of energy could be derived as special forms. In any case, every attempt to form such a concept would fall outside the limits of psychology. In a later connection we shall return to the problem we here leave¹⁵.

10. We have in the foregoing determined thought as psychic energy and sought to show how, provisionally and purely methodically, we must regard it in relation to physical energy. But there will perhaps arise in many the question, *who* it is that *has* or *exercises* this energy? One reasons somewhat thus: "Every activity must be exercised by someone or something. We must distinguish between that which acts and the activity itself. In particular, in the psychological domain we must distinguish between that which thinks, feels, and wills, and thinking, feeling, and willing themselves."

Such a conception, which occurs not only in everyday speech but also among philosophers, spiritualists and materialists in brotherly union, is due partly to the influence of language's mode of expression, partly to the influence of material analogies.

Language distinguishes between subject and predicate and as a rule designates each of them with its own word. There is therefore formed one word for the acting being, another for its activity. Even where it is difficult for intuition to make this distinction between the acting thing and the activity, language nevertheless often employs a formal subject, as in "impersonal" sentences of the form "it is raining." It is a grammatical schema that prevails here, a schema one has no right to give philosophical significance, either psychological or metaphysical. Long ago the acute philologist *Højsgaard*¹⁶ impressed upon us the difference between grammar and philosophy. "Grammatically," he says, "a word or a discourse is considered when one looks not so much at the things that are understood by the words, as at the words themselves, their order and case, and at their mutual dependence or relations to one another. Whereas a philosopher considers only the things that the words call to mind." This warning is constantly needed.

Because language often needs two words to express what happens when something acts, it does not follow that we have two "things" before us. And how, really, should the relation between these two things be thought—the relation between consciousness and that which "has" it, between energy and that which "exercises" it?

In material phenomena, which have exerted precisely the greatest influence on

¹⁵With respect to many details concerning this problem I refer to my *Psykologi*, chaps. II and III.

¹⁶*Methodisk Forsøg til en fuldstændig dansk Syntax*. Copenhagen 1752. (The section "Register og Forklaringer.")

language's mode of expression, we distinguish with ease between the subject of the activity (the movement) and the activity (the movement) itself. We have often seen the body that is now in motion lying apparently still in its place, and we thus know it both as resting and as in motion. When we say that the body moves, it is really these two states that we compare. We therefore fall into embarrassment when we discover that no absolute rest can rightly be assumed. And language especially falls into embarrassment when one has to speak not of "ponderable" but of "imponderable" matter, since this latter is presupposed to be in constant wave-motion. "Ether" can therefore be defined only as that which undulates. It is now the analogy with ponderable matter, and with the apparent difference between equilibrium and motion, that operates when one distinguishes between the soul (the I) and its activity. But the soul we know (like the ether) only as active. What we know here is precisely psychic energy, not any resting "being." The soul's being or nature is activity. It is an intuiting, often a lazy, imagination, not a real thinking, that has the word when one makes that distinction between the thinking thing and the thinking. Spatial images or symbols are confused with psychological concepts. The urge for fixed, resting images, such as material nature seems to present, prevents one from holding fast the concept of activity or energy in its peculiarity and its consistency. Or one fears that the mental life will evaporate if it is not hung upon a "something," a "substance," as upon a secure hook—although one is not so consistent as to ask upon what this hook is fastened. The "substance" could just as well need a "bearer" as the activity. Just as one earlier asked who "has" the energy, so one might here ask who "has" the soul or the I.—Thinkers such as *Fichte*, *Sibbern*, and *Wundt* have the merit of having seen clearly on this point.

We can very well say what we mean by "ourselves" or "our I," without entering into any spiritualistic or materialistic hypothesis. Keeping to experience, we see the mental life unfold itself through different stages and under different forms, and we seek to describe these and to explain the transitions between them. At every given stage of development we then have the mental life before us with certain properties and dispositions, and the task is then to find out what, under certain conditions, can develop from them. (Cf. 8.) "We ourselves" or "our I" has at every stage its content in the properties and dispositions that the phenomena of consciousness make known, and its form in the degree and manner in which these are combined and ordered.

Now the more the continued mental development is conditioned by the given properties and dispositions in their mutual relation, the more energy we ascribe to the mental life and the more active it is said to be. The more, on the other hand, the continued mental

development is conditioned by influences that make themselves felt independently of the individual mental life's past, the more passive it is said to be.

Just as the planets in our world-system tore themselves loose, with a certain initial velocity and initial direction, from the great nebular mass, so the mental life—at the point where it awakens from the night of unconsciousness—begins by acting according to the same laws that it later follows. It begins with peculiar dispositions that give it a certain independence in the face of outer influences. With this agrees the fact that the nervous system is one of the parts of the organism that first appear with distinctness, and that from the first it occupies a central position in relation to other parts of the organism. The functions of the nervous system, to which the mental activities correspond, are not mere effects of outer influence, but determined in form and direction from the individual's first coming into being.—

In what follows we investigate, for the closer elucidation of the nature of human thought, first the more immediate and involuntary way in which it expresses itself in intuiting, association, and comparison (B), then the more conscious and exact form it takes on in judgment (C).

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B. Intuition, Association, and Comparison

a. Sensation

11. When one often distinguishes sharply between sensation and thinking, it is because one conceives sensation as a purely passive coming-into-being of new psychic content, while in thinking one sees a working-over of such content. But one cannot at any point wholly separate passivity and activity—that is, wholly separate states whose causes lie entirely outside an individual's preceding states from such as should be wholly due to these. There always show themselves traces of psychic work, even in the simplest elements of consciousness that can be distinguished. We can be more or less active—that is, the causes of the changes that take place in us can to a greater or lesser degree lie in ourselves, in our character and our preceding experiences; but wholly passive we never are.

If we bring out the conditions for the arising of a single sense-sensation, e.g. a light-sensation, it turns out that the condition for this arising is a change that has a certain degree of strength, a certain degree of suddenness, and a certain degree of

qualitative difference—all in relation to the given state of consciousness. A sensation arises only when the new, in the three respects mentioned, stands in a certain definite relation of opposition to what was previously given. Perhaps, of the three conditions, suddenness is the most important, which from a practical biological point of view is easily understandable, since it is of importance in the struggle for existence to be awakened to attention for changes in the surrounding world, even apart from the degree and character of such changes. But all three conditions point, psychologically regarded, in the same direction. The weaker the change is, the more slowly it sets in, and the more alike the new and the earlier state are, the less possibility there is for the arising of an independent sensation; hence one can rightly designate every sensation as a distinguishing.

It is, to be sure, a distinguishing of a purely elementary kind. The single sensation steps forth in consciousness with its definite degree of independence and with its definite quality, and it is only indirectly, especially by way of physical and physiological investigation, that we can here demonstrate conditions similar to—in any case analogous to—those that condition a distinguishing undertaken with reflection and intent. When we set ourselves to apprehend the difference between two things distinctly, we can do nothing further toward it than to bring these two things into immediate proximity to each other—then the difference announces itself involuntarily, springs forth as it were, since each of the two things stands involuntarily with its own peculiarity. We here produce, by express holding-together, the opposition necessary for a distinguishing; but the apprehension of the difference we call forth only indirectly. And thus too, in the arising of the simplest sensations, the given changes or oppositions are only releasing conditions. The sensation arises as if it were a result of comparison; but the two opposed terms are here not known each by itself; it is only the change in the state, released by the opposition, that appears as an element in consciousness.

It is, then, a justified analogy to call the sensation a distinguishing—and so far a thinking—and already in it to find an expression of psychic energy. Language, which has formed itself under the influence of reflection, has no expressions for functions so simple as those we here confront. Linguistic usage is inclined now to set a very great difference between the simplest psychic functions and the more developed ones, now to read “inferences” and other forms of reflection without further ado into the simplest utterances of consciousness. But by the concept of analogy both the kinship and the distance between sensation and thinking can come into their own.—How one conceives the unconscious basis from which the simplest utterances of conscious life are thus released by the power of the oppositions—whether one declares it to be merely purely

physiological processes, or whether in the unconscious one will see a potential mental life—is in this connection a matter of indifference. The main thing is that we already here trace the chief form of the mental life. For that matter, it shows itself already on a purely physiological consideration how little the sense-sensation can be conceived as a passive effect of stimulations from without. The physical impression traverses, on its way from the outer sense-organ to the great brain, so many layers of nerve-cells that it can only most improperly be said to be the same impression when it reaches the brain as when it comes to the sense-organ. And in the brain itself there is released a reciprocal action of a mass of cells as a consequence of the stimulation. What emerges in consciousness as a sensation corresponds, therefore, physiologically regarded, to a multiplicity of processes; there is gathered into unity in it what physiologically stands as a plurality.

12. That the sensations are thus not simple, passive effects of stimulations from without, but can be regarded as elementary forms of psychic energy, becomes of essential importance with respect to the validity that can be ascribed to them as means to cognition. We build up our world-picture on the basis of them, and yet one became very early aware that they could not be used without further ado to form a coherent and intelligible world-picture. Reflection must at once use them and disregard them if it is to reach its goal. The reason for this is that they are not only the expression of something different from our consciousness, but also of the very nature and mode of operation of our consciousness.

In a characteristic way, reflection's two-sided relation to sensation appears already in one of *Democritus's* fragments, in which the senses speak thus to the understanding: "You poor understanding, you fetch your grounds from us and yet want to overturn us! Thereby you prepare your own downfall."—At the founding of modern natural science this paradox appears again, in that *Galileo*, *Descartes*, and *Hobbes*, at about the same time and on the basis of the same considerations, set up the doctrine of the purely subjective validity of the sense-qualities, all while the new science was nonetheless grounded upon sense-observations. In *Kant* we meet it again in his distinction between matter and form, in that he counts all sense-sensations to the matter and regards only the form—which, however, never occurs separated from the matter—as that with which science has to do.

If sensation were purely passive, it could be assumed simply to reproduce what calls it forth from without. The subjectivity of the sense-qualities is grounded precisely on the fact that they come about as a consequence of an activity on the part of consciousness. That there exists a problem of cognition rests—as will later show itself—precisely on the fact that we are self-active in our cognition. How can we, by our own activity, cognize something different from ourselves? By our very activity we transform it, after

all, according to the laws of our consciousness. Cognition is our work. And in this respect there is, according to the foregoing, no difference of principle between sensation and thinking. The paradox then disappears when one gives up the absolute difference between sensation and thinking, matter and form. The work of reflection then stands as a continuation of the psychic work that sensation has already begun.

Insofar as reflection does the work of sensation over again, it is thus not because with it a wholly new principle comes into force. It is only psychic energy operating in another manner, more appropriate for the solution of the task of cognition. The understanding can thus quite well at once use and overturn sensation. And the “overturning” takes place partly with sensation’s own help, in that the one sense controls the other. To the relation between the qualities within the one sense there can correspond relations between qualities within the other sense, and it is through the analogies that can here be demonstrated that reflection reaches its result. The senses by which the forms of synthesis most clearly come to light, and by which the highest degree of continuity can be attained, are taken as basis. And it is especially the senses that permit clear apprehension of movement. One might call them the rational senses. To these belong sight, hearing, the sense of touch, and especially sensations of movement. The sensation of movement takes place distinctly by a synthesis, since it presupposes the holding-together of the successive positions and stages of the moving thing; only by such a holding-together does movement become an object of consciousness. The special qualities—color, hardness or softness, sound—that correspond to the different stages of the movement could in themselves be thought of as exchanged for others, provided only that the relation between the qualities among themselves continued to be the same. One can then disregard them, and the sensations then become, when they stand in the service of cognition, only symbols for certain stages or places in continuous processes of movement. In a sense-sensation, with its degree of independence and its peculiar quality, a concentration has taken place, a combining of what, physically and physiologically regarded, is a multiplicity. Modern physical theories interpret what for us is only the object of a single act of sense as an aggregate of small parts in constant movement. Our sensation appears here as analogous to a statistical result of a mass of particulars that are each by itself inaccessible to us and that stand in the most complicated mutual connection. The kinetic theory of gases is especially an example of this. According to it, every gas is an elastic fluid whose elements are in constant movement, with numerous collisions against one another and against the barriers, and consequent changes of direction and velocity. For our sensation it nonetheless appears as a whole, within which we distinguish nothing. And the same

holds, as already remarked earlier, of the relation of sensations to the brain-processes. What in our consciousness appears as a single sensation corresponds to a very composite reciprocal action between the millions of cells of the great brain and between these and the other parts of the brain.

What takes place in the nervous system and in the brain does not resemble what stimulates the sense-organ from without. And what takes place in the brain probably does not resemble what takes place in the sense-organ, the ingoing nerve, and the lower centers that the impressions pass through on their way. The sense-sensation resembles neither any of these physiological processes nor the outer stimulations that release them.

Innumerable enigmas conceal themselves here. And yet it is a fact that sense-sensations can guide living beings in their struggle for existence. Our states can very well be decisively determined by our own nature, provided only that they lead to the release of activities by which we can maintain ourselves against the surrounding world. In instinct, which is released by definite sense-stimulations, this appears especially clearly. The answer is given in a quite different way than the question is put. In instinct, urge and capacity are so united that they are both released and operate together when a certain definite change sets in in the state of the living being. Thus when the pregnant bird builds a nest, or when the newly hatched chick scratches if it feels gravel under its feet.

Already the first thinkers who became clear about the subjective nature of the sense-qualities emphasized, in connection therewith, the practical significance of sensations as the original, the theoretical significance as the derived. Thus especially *Descartes*, *Malebranche*, and *Berkeley*. Modern evolutionary theory has again brought out this point of view. Sensation is first and foremost a means in the struggle for life, only secondarily a means to come to know the nature of existence. The connection between its practical and its theoretical significance lies in the concept of sign or symbol. The sign can be a signal for action; but it can also become a signal that announces the setting-in of a certain change, apart from the action necessitated by this change. There are many intermediate degrees between these two significations. But in neither of the two significations is there needed any likeness between the sign and what is to be signified. And the sign perhaps operates best by being as simple as possible—by presenting as a unity what is in itself a multiplicity. The main thing is that it can serve as an introduction either to an action or to a train of thought that can attest their validity—the action by leading to the satisfaction of an urge, the train of thought by leading to understanding and to agreement with new sensations. In this last case too it is a matter of the satisfaction of an urge—namely of the very urge to understanding, to the agreement of our reflection with

as many sensations as possible. Motive (value), form, and object thus always operate together. (Cf. 6.)

The special qualities with which the different sense-sensations arise nevertheless contain a great, perhaps insoluble, problem. For from a biological as well as from an epistemological point of view, whatever quality a sensation appears with, sensation can perform its function in the service of life and of reflection, provided only that it is able to release activities or trains of thought that can satisfy the practical or theoretical need. Why precisely these definite colors arise for us when certain electromagnetic stimulations have propagated themselves through space to our eye, or why we hear precisely these definite tones when certain air-waves have propagated themselves to our ear, stands as a constant enigma. We stand here at mere facts and must for the present remain standing at them.

Each single one of our senses has its series of qualities, more or less continuous and systematic. Even where the continuity is greatest, as in the domain of the sense of sight, it is nonetheless interrupted at definite places, and the series of qualities is always limited. To infra-red rays the retina is not receptive, and ultra-violet rays affect it only at a particularly great degree of strength. This limitation, which probably has its ground in certain physical or physiological conditions, and perhaps has its appropriateness, entails a limitation of the possibilities of knowledge.

Each sense-organ has its own peculiar advantage. The ear possesses the fine power of discrimination by which partial tones can be recognized within the sound in which they occur. The eye's power of discrimination shows itself especially as the capacity to localize the different visual impressions. In the different living beings, according to their conditions of life, different senses show themselves especially developed. All such special degrees of development must, just like the limitation, be assumed to have arisen through reciprocal action between the inner and the outer conditions of life.

When we here speak of living beings and of a surrounding world, we make use of a knowledge that, like all other knowledge, is itself won on the basis of sensations and by the work of reflection on this basis. It is then presupposed that the psychic energy that already expresses itself in sensation has developed in the form of reflection. The biological knowledge we make use of for understanding the development of the psychic activities is itself a product of these activities.

13. It has already shown itself above that a single sensation is an abstraction. The character of every sensation rests on the relation between the conditions under which it arises and the conditions under which simultaneous or preceding sensations arise. And

there showed itself here an analogy between its arising and the discovery of a difference effected by reflection. It was for this reason that we could call it a distinguishing.

The psychic equilibrium must be abolished for a sensation to arise. But there always expresses itself a tendency to equalize this disturbance. And this tendency appears in two forms, a more elementary one, which consists in fusion, and a higher one, which consists in the formation of a sense-intuition.

Fusion sets in, as already indicated, when the change produced by the impression is slight in relation to the state we are already in, or when changes follow very rapidly upon one another. Examples of fusion we have in the timbre of tones and in the cooperation of colored and uncolored light-impressions in the coming-into-being of light-sensations. In a visual image due to seeing with one eye, it is not noticed that the blind spot on the retina gives no contribution to the filling-out of the image. When a regular alternation takes place of light-stimulation in the time a and complete non-stimulation in the time b, there is nonetheless obtained, when the time $a + b$ is very short, a continuous sensation. Even changes that take place simultaneously in different sense-domains fuse insofar as they mutually inhibit or reinforce one another.

It would scarcely be right to see, in such a fusion—in which the individual elements cannot make themselves independently felt—the original form of conscious life, so that an “undifferentiated continuum” should be the starting-point for mental development¹⁷. All conscious life is scarcely ever fused; there will in any case probably stand different fused complexes over against one another. A complete state of fusedness would be a state of rest, which would be one with a state of unconsciousness. A certain breaking of oppositions, or at least of differences, is necessary for the subsistence of conscious life.

When fusion is not possible, and there nonetheless expresses itself a tendency to equalization, a sense-intuition can form itself, in that the sensations are ordered in such a way that each single one can make itself felt without swallowing up or supplanting the others. With the least possible change, the new content that a sensation brings is then incorporated into the whole connection of consciousness. The sense-intuition solves a double task, in that it at once lets the novelty and peculiarity of the single sensation and the connection of conscious life come into their own. It presents at once multiplicity and unity, while fusion abolished the multiplicity. The sense-intuition is clearest when the differences have previously appeared each by itself and only thereupon are connected in such a way that a single act of attention can survey them. The clearest form of such a surveying is space-intuition, which we therefore have a pronounced tendency to use as a

¹⁷Einar Buch: *Om Fornemmelers Sammensmeltning, særlig ved Klangindtryk*. Copenhagen 1898, pp. 31–35; 60 f.

schema for everything we wish to make clear to ourselves. In particular, a time-intuition first becomes possible through the symbolic use of space-intuition, although we can of course quite well apprehend change and succession—e.g. of tones or of our own inner states—without using a spatial schema.

Sense-intuition denotes a higher stage of psychic energy than fusion. The multiplicity is preserved through an involuntary ordering. There will always be one or more single elements that determine the ordering, and around which the others group themselves—namely those that correspond to the dominant urge or interest. Therefore every involuntary sense-intuition will be articulated, pointed in a definite direction, that in which attention is concentrated. This determinacy of direction forms an analogy to what later stands as purpose for striving and object for reflection. Out of this concentration the other members in the sense-intuition fade off into the indefinite and obscure.—

In this description we must constantly remember what was above called the psychic antinomy. When it is said that elements fuse or become members in a sense-intuition, the meaning is not that the elements, apart from these kinds of synthesis, have the same character as they have within the syntheses. It is only in our abstract language that we can let the elements go in and out of the different connections like a kind of atoms.

b. Recognition

14. A pure sense-intuition would be one in which a multiplicity is involuntarily ordered without earlier lived experiences or experiences contributing. Apart from the most primitive states, such a pure sensation scarcely occurs. As soon as, in the multiplicity that is ordered, there are found elements that have previously been present, there is the possibility of a new form of psychic energy: recognition. The simplest example of recognition we have where a sensation, by reason of repetition, immediately appears with a peculiar quality, different from that with which it appeared when it first appeared. I use the expression quality to designate the wholly immediate difference that can obtain between the known and the new—a difference that can be as simple and indescribable as the difference between yellow and blue. The quality that a sensation can acquire through repetition I call the familiarity-quality.

While in the simple sense-intuition it is essentially the difference between the elements that operates, in recognition the relation of likeness makes itself felt. What conditions the transition from new to known is not itself something new, but is, on the contrary, precisely a repetition. How the familiarity-quality is to be explained physiolog-

ically is not decisive for its psychological and epistemological character. It is, however, most natural here to point to the influence that repetition and practice have in altering the course of processes and functions. Through repetition one and the same function, probably also the brain's function, can take place with greater ease, and recognition might then be the psychic correlate of the altered character of the functions. However this may be, the circumstance that the relation of likeness now makes itself felt marks a significant advance. One can already here speak of an understanding: the present here stands immediately as one with something earlier experienced. There is a foothold, a home for it in consciousness. *Plato* called the dog a philosophical animal, because its behavior toward men is determined by whether it knows them or not. In this sense newly hatched chicks too are philosophical animals, since they scratch only when they feel gravel under their feet, not on a carpet. The remarkable thing about instinctive actions is precisely that they are released by stimulations that are at first new to the individual and yet operate as if they were known.

In everyday speech we often use the word "understand" in the sense of recognizing. It is an understanding of the most elementary kind. I understand a sound when I know what kind of sound it is, that is, recognize it. I can in this respect understand a word when I know its sound, even if I do not know its meaning. Understanding has many degrees. I understand what kind of figure it is that approaches when I recognize the human figure in it; but there can still be much to understand, namely which person it is, and so on¹⁸—

Just as in the simplest apprehension of difference, so in the simplest recognition language lacks apt designations. But there is a significant analogy between such recognition and the comparison undertaken with reflection. In genuine, free comparison two independently given sensations or representations are presupposed, whereas immediate recognition takes place without the present and the earlier experienced needing to appear each by itself in consciousness. Just as a sensation can arise immediately when there is a certain opposition between the influence of a stimulation and the state at hand, so immediate recognition arises on the presupposition of a relation of likeness between the influence of a stimulation and the dispositions or after-effects that earlier stimulations have left behind. In both cases it is only the results, not the presuppositions, that appear in consciousness. The elementary distinguishing and the immediate identifying are the first intimations of what at higher stages appears as thinking in the narrower sense of

¹⁸It is therefore not an apt objection that *Husserl* (*Logische Untersuchungen*, II, p. 73 f.) makes against me, that unintelligible words too can sound familiar. In the wholly immediate "understanding" of a word, it is only the sound or the sign, not the meaning, that is known.

the word.

15. Distinguishing and recognizing denote two poles in our cognition, which under different forms are found again at all its stages of development, and which correspond to two necessities of life: receptivity for the new, the changing, and the capacity to take it up into oneself in as intimate a way as possible. The two capacities can stand in sharp opposition to each other. They are two different kinds of psychic energy, a centrifugal and a centripetal one. But there is a natural reciprocal action between them. Recognition makes possible a saving of force. If, in the face of changes that repeat themselves, we had to go through anew each time the experiences that the first change called forth, it would lay claim to all our energy, and we would not get anywhere. By reason of recognition, on the other hand, we can turn with fresh force toward the new, and the very opposition between the known and the new lets the new, by a contrast-effect, step forth more vividly and characteristically.

Yet recognition itself presupposes energy to perform the synthesis of the present and the earlier. In that weakening of psychic energy which *Pierre Janet* calls psychasthenia, and which essentially expresses itself as a lack of will (aboulia), the capacity to recognize (understand) the present situation is often lacking. The bond that should connect the present with the earlier is broken. On the occasion of one of his aboulie patients, Janet says: "The attention to the present, the connection (synthèse) of the new sensations with the old images, which constitutes recognition, has here, like the will, vanished on account of the mental weakening"¹⁹. Yet such weakening can also lead to false recognition or memory-illusion; for energy is also required to distinguish the new from the known.—

It is only an analogy that obtains between immediate recognition and reflection. One has no right to identify it without further ado with acts of thought performed with clear consciousness. It would thus be incorrect to call it a judgment, as several authors have done²⁰. For a judgment there is required a transition of thought from a subject to a predicate; but in immediate recognition no transition takes place between two terms; the familiarity-quality appears as if at a single stroke. Immediate recognition can form the basis and presupposition for a judgment, but the judgment first comes into being when a distinction is made between the given, present thing and the earlier thing with which it is identified. The so-called exclamatory judgments, which we shall later come to discuss, are the first judgments that become possible on the basis of recognition; in them there is a predicate, even if there is no subject. Thus when the gypsy boy in *Blicher*

¹⁹*Les Obsessions et la Psychasthénie*. II, p. 29.

²⁰*Meinong: Beiträge zur Theorie der psychologischen Analyse*. p. 374.—*Hans Cornelius: Versuch einer Theorie der Existenzurteile*. p. 56 f.

cries: "A dog! A hunter!" These are immediate expressions of recognitions.

16. When it is a whole, not a single property or a composite phenomenon, that is recognized, it will not always be necessary to recognize expressly all the members (the parts or properties). Recognition of a single member can be sufficient, and the familiarity-quality is then involuntarily transferred to the other members, so that the whole stands as familiar, or an expectation is awakened that the other members too are present—an expectation that can be so secure that one does not seek confirmation of it. The recognition is then partial, supporting itself merely on a part of the whole that serves as a mark of recognition.

If the expectation leads to a continued going-through of the members of the whole, and these, when they are sensed, correspond to the expectation, a mediate recognition of them takes place. A person's gait and bearing seem familiar to us, and we then expect that other properties will correspond to them; we address him, and his expression and speech are as we expected; they are then recognized mediately, in that the representation contained in the expectation covers what we see and hear.

In immediate, in partial, and in mediate recognition alike, an involuntary interpretation is woven into apparently simple observations. It is a surplus of energy that makes us at once interpret the given and, in virtue of the fundamental law of synthesis, bring it into connection with previously given content. Later, renewed observation may perhaps correct our first interpretations and expectations. But it is only by the contribution we ourselves give that we really come to know things. If we sat still and waited, the circle of experiences would become narrow. Now, on the contrary, the urge to use our energy leads us to an involuntary experimenting, which can then confirm or correct our expectations and representations. We have here, in purely elementary form, already an intimation of the method that developed science applies: an observation (recognition) leads to a conjecture (expectation) or a hypothesis, which can be confirmed or corrected by new observations. In the simplest mediate recognition there is already described that arc from observation through reflection to new observation in which the peculiarity of all scientific method consists. This kind of recognition testifies to greater psychic energy than immediate recognition: the latter is bound to the given and present, whereas the former goes out beyond this in expectation and representation. It is from the first the urge that drives out beyond the given—partly the urge for an outlet for energy, partly the urge for the attainment of something that does not involuntarily fall to our lot.—

Some psychologists have held that there is no immediate recognition, but that all recognition takes place mediately, that is, through previously awakened expectations and

representations, and perhaps even through express comparison. After the discussions conducted on this, it can nonetheless be regarded as settled that there is an immediate recognition, and that it is an interesting transitional form between sensation and reflection²¹.

c. Memory- and Imagination-Intuition

17. The longer the interval becomes between the arising of the expectation and its confirmation or correction, the more independently the representations appear—the more liberated they become, or can become, from the impressions that first awakened them. There can then—on the presupposition of sufficient energy—form itself a content of consciousness with a certain independence of the supply from without, a circle of free representations.

Just as little as the sensations do the representations subsist independently of one another. They are members in a train of representations. They call one another forth and connect themselves with one another, and by reason of their mutual connection (association) a new kind of intuition becomes possible, *memory- and imagination-intuition*. In many cases one can fairly distinctly point out the process of association by which such an intuition forms itself; but the formation can also take place so rapidly that a total image appears in consciousness at a single stroke. This holds especially in inspired imagining, where nature operates “as if it were art,” performs deeds that in other cases would require conscious deliberation, if they could be performed at all. Only by roundabout ways can one, in inspired production, gain an insight into how imagination-intuition has come about. And the matter becomes still more complicated by the fact that urge and interest cooperate in all memory and imagination, and that this cooperation here takes place more hidden than in sensation and recognition. The whole past of conscious life operates along with it. The psychic energy is therefore of a higher degree than in the more elementary processes.

Memory- and imagination-intuition is, like sense-intuition, a form in which the energy of consciousness solves the task of combining a multiplicity of elements. Both the unity and the multiplicity are to come into their own. In memory and imagination a more or less rich content of experience and representation is worked together into a

²¹Cf. my *Psykologiske Undersøgelser* (Vid. Selsk. Skr. 6. Række), pp. 8–26. Katz in *Zeitschr. für Psychologie und Physiologie der Sinnesorgane* XL., 6. 1906. p. 432. Herrlin: *Minnet*. pp. 128–218.—H. Bergson nonetheless presupposes, in his critique of the theory of an immediate recognition (*Matière et Mémoire*. p. 91 f.), that all recognition presupposes a free representation.—Besides the kinds of recognition discussed above (immediate, partial, and mediate) there are several kinds. In *Psykologiske Undersøgelser*, p. 35 f., I have attempted a systematic survey.

whole, often in such a way that a quite different ordering comes about than the original one. Such total images have their definite directions, determined by the interests or purposes that cooperate in their formation. Every time we lay a plan, we make use of the past to prepare the future and transform, often without ourselves knowing it, the given representations in such a way that they can enter as members into the new whole. When it is intellectual or aesthetic interest that determines the ordering, it is not an outer end that is aimed at, but the involuntary endeavor is directed at the intuition's becoming as clear or as characteristic as possible.

Once an intuition has been formed, there will be a striving to hold it fast as long as possible. Under the influence of new experiences, incitements, or associations of representations, individual members are perhaps shed or taken up, or else the mutual relation of the members is altered. The direction of the intuition can also be changed. Just as in quite simple organic beings, which have no proper limbs, displacements of the organic mass in certain directions can take place, so too the intuition can be articulated functionally without being replaced by analyzing and judgment. A simple example from the domain of sense-intuition is offered by the so-called Schröder staircase figure, which by changing accessory sensations and associations of representations, or by altered attention, is seen now as a staircase, now as an overhanging wall. When we stare long down into the water from a bridge, it can come to look as if we ourselves and the bridge were moving in the direction opposite to the water's real movement. According to changing moods or outer occasions, an image that stands in our memory or our imagination can thus shift without our noticing it. It can be one of our earlier experiences that is thus altered, or the image we have formed of a poetic figure. Our memories and imagination-images have their own history, whose individual stages it is not always easy to trace, any more than it is easy to find the grounds for their shifts. Their history takes place in the involuntary mental life, right down in the ground of nature within us, and reflection can here often only ascertain that changes have taken place, when it holds together two mutually divergent intuitive images of the same object from different points of time in our life. In other cases reflection can in time become aware of the change that threatens to set in, and work to prevent it. The astronomical world-pictures, which from the first formed themselves involuntarily, are an example of this. Scientific hypotheses often have their advantage in their intuitability and in the support they have in sensation's apparently immediate testimony, and there is therefore often a hard struggle before they are let go. To a still higher degree this holds of religious intuitions, in which a whole host of life-experiences—experiences of the course and value

of existence—have formed themselves into great images. Here too the most important thing—both as regards the intuition’s original coming-into-being and its alterations—takes place right down in the involuntary mental life, although reflection can enter the lists for the right to hold fast, despite everything, the images that have been laid down. In the relation between intuition and reflection many individual differences make themselves felt. In some it is feeling that holds back, and reflection is then forced to fight for the images in which it has found rest. In others it is the urge for intuitability that opposes reflection’s intervention²². In others again there is such spiritual elasticity that emotional interest and the urge for intuiting relatively easily give way to reflection.

In animals the capacity to help oneself with functionally articulated sense-intuition probably has preponderance over the capacity for reflection, if the latter occurs at all except in the most highly developed animals. The dog that runs toward a man whom it “believes” to be its master will, during the running, constantly alter its intuiting, according as the man’s features become more distinct, and when these features finally show themselves to be quite other than the master’s, a disappointment will arise; but whether the free memory-image of the master thereby appears in its opposition to the sense-image of the stranger is impossible to decide. There is no reason to assume a proper judgment or inference.

What I call articulated intuiting, or functional articulation of the intuition, is nonetheless regarded by several authors as a judgment. But it is then in any case a judgment of a quite different kind than the proper judgments that rest on reflection. When *Gassendi* thus ascribes to the dog that “believes” it meets its master a capacity to judge, he adds: “An affirmative judgment is, after all, nothing other than the apprehension of a thing with an addition or a property.” The dog sees, *Gassendi* holds, the man immediately as its master, and this seeing or representing (*imaginari hominem herum*) is—according to *Gassendi*—the same as forming the judgment that the man is the master (*enunciare hominem esse herum*); yet this judgment is passed tacitly or potentially (*tacite vel virtute*)²³. In a similar way several psychologists express themselves. *Lloyd Morgan* thus distinguishes between perceptive and conceptive judgments. The perceptive judgment (perceptual judgment) is what I call articulated intuiting, in that it consists in a single feature’s becoming predominant in a previously formed image. A conceptive judgment, on the other hand, presupposes a conscious distinguishing between individual features; only this, therefore, is a proper judgment. What *Morgan* calls a perceptive judgment, *Ribot* calls the logic of images (*logique des images*). This expression is more fortunate,

²²Cf. on the opposition between intuitive and reflective natures my *Religionsfilosofi* § 95.

²³*Gassendi: Opera*. Lugd. 1658. II. p. 411.

since the image-forming, intuiting process is decidedly the essential thing in the phenomena here in question. They belong under intuition, not under judgment²⁴. The expression articulated intuition seems to fit best, since the intuitive image need not be dissolved because a single element is strongly brought out. If a new articulation within the intuition is not possible, a new intuition can form itself, within which the changes come into their own. Possibly the earlier intuition then vanishes entirely from consciousness. If, on the other hand, it is held fast or reproduced, a comparison and an analysis can get going, which may perhaps lead to a judgment.

The articulation stands as a preparation for, or a presage of, an analyzing and judging. Thus what at later stages of development becomes a special organ is, in embryonic development, intimated in bulging parts of the still uniform and continuous tissue.

18. In itself one might suppose that the intuition-formations, with their shifts and articulations, could be continued to infinity, without any judging conditioned by reflection being necessary. It must be taken as a simple fact that this is not so, and that—at least in human beings—a necessity for reflection and judgment sets in. The equalization and equilibrium that can be attained through intuiting does not always show itself to be sufficient. Conditions can arise that make the combining have to take place with clearer consciousness than happens in the involuntary and half or wholly unconscious process by which the intuition comes into being. In sense-intuition, recognition, memory- and imagination-intuition, it is really only the result that appears clearly in consciousness, while the formation-process itself takes place predominantly below the threshold of consciousness.

Reflection is in a certain sense not productive. It only brings to clear consciousness what is already contained in the involuntarily formed intuitions. It compares and tests, chooses and rejects, connects and separates, fills out deficiencies and determines exactly the mutual relation of the elements. The new that reflection brings and expresses in concepts, judgments, and inferences is the clear consciousness of the individual parts of the content of consciousness and of their mutual relation. Through the purgatory of reflection it can soon show itself that we know more than we thought we knew, in that there can lie in the content of consciousness unsuspected presuppositions for orderings and inferences—soon that we know less than we thought we knew, in that incompatible elements or invalid connections are discovered.

²⁴Lloyd Morgan: *Animal Life and Intelligence*. p. 328 ff.—Th. Ribot: *L'évolution des idées générales*. p. 32 ff.—Hobhouse (*Mind in Evolution*. p. 200) ascribes to higher animals a capacity for "practical judging." But he adds cautiously that if animals have no proper [free] representations, they must have something that can perform a corresponding function.

Judgment is the form in which the content of consciousness can come into equilibrium anew, when reflection has caused a dissolution of the intuition. In a judgment different elements of consciousness are brought into definite and conscious connection with one another. Insofar as the appearance of judgment is a sign that the wholeness and equilibrium of the intuition no longer subsists, it can be regarded as a sign of an imperfection, especially since the single judgment cannot express the whole fullness that can lie in an intuition. Perhaps an infinity of judgments is required for the whole content of consciousness, with all its nuances, to come into its own. *Goethe* thus sets “das zersplitternde Urteil” in opposition to “die stille Fruchtbarkeit” in immediate intuiting, and *Wundt* even defines judgment as the dissolution of a composite representation. But judgment in reality signifies a new formation of a whole, which has become necessary because the tension within an involuntarily formed intuitive whole has become so strong that a dissolution must set in. It has a similar advantage over intuition as intuition has over fusion: both the differences and the likenesses in the content step forth more distinctly than before. The formation of judgment is therefore a higher expression of psychic energy.

There must of course be a close connection between the judgment and the intuition it builds on. One often expresses this by saying that the judgment “lies in” the intuition, or that the intuition is a potential judgment, as *Gassendi* said. The difference between intuiting and judgment is that the latter lets the elements step forth independently, so that a distinction can be made between “subject” and “predicate,” while in the intuition a more continuous connection prevails. When I believe I know a person who approaches, I see him immediately before me as the one I take him to be; but if I feel an urge for a closer testing, I institute a comparison that distinguishes between the figure that approaches and the memory-image of my acquaintance, and thereafter there is the possibility of a positive or a negative judgment.

But the way from intuition to judgment can be long and toilsome, and it is not said that the “possible” judgments can become actual. There can be a logical possibility for a judgment without a psychological possibility being present. Thus it is said to *Peer Gynt*: “We are thoughts; you should have thought us!”—The logical possibility is present when the intuition contains elements that can be separated out, and whose mutual relation can receive a closer determination. But this separating-out and determination is undertaken only when there are sufficient conditions for an abolition of the whole equilibrium of consciousness that the intuition denotes. Here, as at the first arising of consciousness, there must take place an interruption of the uniform state, in order for a further-reaching

mental work to get going. A special interest must attach itself to individual elements, so that an urge arises to become clearly conscious of their mutual relation. Thus, e.g., if an intuitive image contains something that could become a means to the attainment of a purpose, or serve for a better understanding of another intuitive image, or form a contrast to it.

The judgment must correspond to the intuition, since it is to express its content in another way, more appropriate under the given conditions. But this does not exclude that it is an essential change that takes place in the transition from intuition to judgment. When one does not embrace the psychological atomism that I have combated above, it must be clear that the elements of consciousness undergo essential changes by passing into new connections, and the connections of judgment can be quite other than those of intuition. The elements are not indifferent to the connection in which they stand. In the intuition an element stands in a single special connection; the judgment brings it into connection partly with elements that lie within the same intuition, but perhaps at great distance, partly with elements from other intuitions. In the intuition I dwell, e.g., on a man's action as a total image and am absorbed in the mood this awakens in me; in the judgment "he has acted nobly!" I make use of a standard with which I more or less consciously compare the action. The validity of the judgment presupposes only that from the relation established between subject and predicate one can infer back to the intuition that lies at the basis, so that the judgment can again lead to a construction of a similar intuition in those who hear the judgment without having had the intuition at first hand. There must be analogy, but not necessarily identity, between judgment and intuition.

d. Association of Ideas

19. The content to which judgment corresponds does not always appear in the resting form of intuition, and as a surveyable whole. Judgment can be formed out of a series of associations of representations, in which intuitive images of greater or lesser extent replace one another more or less rapidly. The difference between association and intuiting is only a difference of degree. The more the successive gains the upper hand over the simultaneous, the more association—which in its higher degrees becomes a flight of thought—steps in place of intuition. What intuition is in static form, association is in dynamic form. Already in the formation of memory- and thought-intuitions associations cooperate, and no less is this the case in shifts and articulations, as well as in the dissolution of the intuition.—But we now regard association as an independent form of

the involuntary conscious life.

Just as involuntarily as an intuitive image can emerge in consciousness, just as involuntarily can one representation draw another after it, this a third, and so on. In both cases laws operate that do not themselves come to consciousness. The articulating intuition stands closest to association; but in it too the dynamic is still subordinate to the static, since the shift quickly leads to a resting image.

The transition from association to judgment takes place through the relation between the members of the association-series becoming the object of distinct consciousness. The urge for this will already arise from practical interest, as when the preceding members stand as means in relation to the following ones. Where merely practical interest prevails, one will attend to the preceding members only insofar as they indicate means to the attainment of an end, and the whole order in which the members are brought will be determined no longer by their involuntary sequence, but by their relation to the end. It is, as *Hobbes* remarks, the thought of a purpose (*cogitatio finis*) that makes us, in the waking state, order our representations in a definite and coherent way, while in dreams, where the firm thought of purpose is lacking, they can unfold more chaotically. For a thoroughgoing ordering there will be required holding-together and comparison with other association-series and intuitions than those involuntarily setting in, and the more such series and images thus interact, the greater the psychic energy that comes to light.

Every attempt to order, by way of reflection, the members in the involuntarily arisen association-series consists in a testing of the members' relations of likeness and difference. Even if the objects *a* and *b* have occurred together ever so often in my experience, so that *a* has an irresistible tendency to call forth the representation of *b*, the justification of expecting *b* when *a* has set in will nonetheless depend on whether the *a* that earlier occurred as predecessor of *b* is identical with the *a* that has now set in, and further on whether it can be shown that *a* cannot set in without being followed by *b*. The greater the continuity between *a* and *b*, so that they really form a whole, the more justified the expectation will be. If it is an association by likeness that we have before us, the transition from the one member (a_1) to the other (a_2) will be the more justified the more the likeness approaches identity; it is then merely a transition from one example to another. Thus association approaches judgment the more the transition from object to object is determined by continuity and identity. In the genius, the formation of judgment can proceed as if it were an involuntary association-process, since the psychic energy is released with great ease and in great fullness. Reflection is here limited to a minimum. In everyone, practice in a definite train of thought can operate in the same direction.

Where reflection can be spared, it is spared.

20. Association of representations is, just as much as reflection, an expression of psychic energy, and it is wrongly that one has regarded it as a purely passive process and set it in absolute opposition to “thinking.” What happens in every association is really that a whole, a larger connection, is built up, constructed out of a single given object that occurs in a certain isolation. There are indeed always several objects or elements that cooperate at the springing-forth of an association-process; but a single one will be the decisive one, and for simplicity’s sake we can keep to it. Consciousness has both an urge and a capacity to go out beyond the single object or element and connect it with others. In this instinctive tendency the psychic energy expresses itself. All association can be traced back to a transition from part to whole; what we call association by likeness and by contiguity are only limiting cases. In every association relations of likeness and of contiguity operate together, in such a way that now the one, now the other has preponderance²⁵. The urge for the release of energy is, in the domain of the life of representation, so strong that a minimal occasion can operate as a release. There is here an analogy to the urge for movement. Now the incompletely given is filled out (by predominant association by contiguity), now one circles around it through dwelling with kindred representations (by predominant association by likeness). In both ways the single element can be incorporated into the whole psychic world, and its relative isolation can be abolished. Which of the different paths is taken in the particular case—and which special relations of likeness or contiguity come to operate—nonetheless does not depend on purely intellectual conditions. Like a hidden weight, interest can press the scale-pan down for a single definite relation of likeness or contiguity that in itself would not be decisive. It is ultimately man’s striving that determines his associations of representations, even though these, once they have got going, can in turn alter the strength and direction of the original striving.

When now association is replaced by judgment, the relation is reversed. It is no longer a transition from part to whole that takes place, but a transition from whole to part, that is, an analysis. Association is, like intuition, an integration, but judgment presupposes a differentiation. In the association-process it holds that a little tussock can upset a great load; in the formation of judgment a needle is found in the great load. The associations can bring forth the whole content of consciousness, but the formation of judgment draws out a single thread from the rich web. Association is productive, the formation of judgment is critical.

²⁵See more closely my *Psychologi* V B., 8.

Yet neither is this opposition complete. The associations are—as already remarked—not determined exclusively by a single element. This is seen already from the fact that one and the same representation does not, under all conditions, awaken the same associations in one and the same individual. And on the other hand the formation of judgment presupposes that attention is directed toward a single element's relation to other elements (in or outside one and the same association-series), and so far it too can be said to begin at a single point, with the little tussock.

Yet another opposition has been found between association and judgment, in that the former is supposed always to be able to be continued, since there is no inner limitation for it, while the latter is supposed to denote a definite conclusion. "Association," says *W. Jerusalem*²⁶, "is never concluded in itself. Ever new members attach themselves, ever new deviations from the original direction take place, and nowhere is there found a natural conclusion lying in the very nature of association ... In judgment a process is released from its connection with the rest of the content of consciousness; it is isolated and, as it were, concluded for consciousness. We are for the present done with it, and precisely this concluding moment in the act of judgment forbids regarding it as an association." Against this it must be objected that an association-series can very well find its natural conclusion, namely when the motive that has called it forth is fully satisfied. One can also reach a representation that perhaps lies far from the starting-point, but that offers an independent interest and therefore takes attention captive. When one takes into account the interest, which is never lacking in the association of representations, an inner conclusion of this can very well be given, not to mention that the relations of likeness and contiguity that are at one's disposal in the particular case will always be limited. On the other hand, judgment does indeed denote a settling of accounts, a decision conditioned by reflection, but only a provisional decision, since new reflection can correct it or else bring it into connection with other judgments and thereby unfold what "lies in" it.

e. Comparison

21. The transition from intuition or association to judgment takes place through reflection, which consists in an analysis, a successive turning of attention to the individual objects and elements in the intuition or in the association-series. Thereby free comparison becomes possible, in distinction from the elementary comparison in sensation and the bound comparison in recognition.

²⁶*Die Urteilsfunktion.* p. 79; 82.

There lie two objects before me on the table, which I can see at a single glance. But although they are both within my field of vision, I can fix my attention now on the one, now on the other, move back and forth between them without necessarily moving my eyes. The one object does not fall out of the field of vision because attention is directed toward the other. By such alternation of attention an experiment is made, whose result is that the two objects stand either as similar or as different. The likeness or the difference springs forth for us as a peculiar quality, an element that was not present as long as we regarded the objects each by itself without holding them together. It is this holding-together that brings it about that the new property we call likeness or difference comes forth. The older English psychology (especially *James Mill* and *Stuart Mill*), which laid the whole weight on the individual representations each by itself, could not explain how comparison was possible. For them, relation-representations differed from other representations only in that they presupposed two “things” instead of one.

The active element in comparison—apart from the energy that holds together the elements that are compared—is one with attention. This is a successive concentration toward the different members of the intuition or the association, a concentration that precisely by reason of its alternation lets the content between the elements step forth distinctly. Experience shows that we, all else being equal, apprehend a successive relation more distinctly than a simultaneous one.

If the objects we compare are not given simultaneously, the comparison comes about through the one member being reproduced while the other is directly observed. Memory thus provides the simultaneity within which attention can then swing.

If, at the passage of attention from the one object to the other, no difference at all appears—apart from the exertion involved in the alternation of attention itself—the relation between the two objects stands as a *relation of identity*. Identity in the strictest sense is the likeness an object has with itself. When different objects are said to be identical, the meaning is that it is as if attention returned to the same object; and conversely, that an object is the same means that attention can return to it without any difference appearing.

If it is emphasized that the two objects, despite their lack of difference, nonetheless occur at different times or in different places or both, their relation is called *Sameness* (*Congruence*).

If some of the two objects’ parts or properties appear as identical or the same, others as different, we get a composite *Likeness*. It has sometimes rightly been held that all likeness—apart from identity—was of this kind.

If a likeness is present without there being identity or sameness at any point, we can call the relation *Qualitative Likeness*. It is a peculiar kind of qualitative difference. The remotest degree of likeness is *Relational Likeness (Analogy)*, where the likeness does not concern any single property or part in the two objects, but the relation between properties or parts in the two objects.

22. Comparison can take place involuntarily or voluntarily. Comparison is involuntary when no intention, no preceding representation, no wish to find out likeness or difference makes itself felt. Voluntary comparison presupposes that one has set oneself as a task, as a purpose, to find whether there is likeness or difference between two objects. The transition from involuntary to voluntary comparison is formed by such cases in which the one object is reproduced, so that it is merely held together with the other. It is a transition reminiscent of the transition from immediate to mediate recognition. In mediate recognition a representation is released, which thereafter covers a following sensation. An analogy in the domain of will is formed by the transition from urge or instinct to drive (when by drive we understand an urge connected with a representation of its goal). And here there is really more than analogy; for there is an urge to compare just as to act, and in both cases it can be of importance that one becomes conscious of what one feels an urge for.

Now it is not always the case that both the objects to be compared are given or can without further ado be called forth in memory. Often only the one object is at one's disposal, and the task is to find another object that stands in a definite relation of likeness or difference to it. The more clearly and distinctly this relation can be represented, the greater the possibility that the new object can be found. The most exact indication of that relation we have where the task can be put into an equation. We then have a definite standard to decide whether the sought object appears or not, and by a methodical procedure we can perhaps force out the object that satisfies the demand. In a more indefinite way something similar happens when we wish to call forth something in our memory and know of it only that it stands in a certain connection with a representation already present in us; by concentration of attention on this, or, as it has been called, by a boring, the sought thing can then spring forth. The psychic energy appears quite especially in such cases. The point then is not merely to connect given members, but to find a member that is perhaps quite new, but that stands in a familiar relation to a given member. Where we can indeed determine the relation to the given member, but on account of the limitation of our experience cannot find the new member, there we end in a problem. Thus we are placed, e.g., when we assume that in the psychic domain there is

something corresponding to the difference between potential and actual energy in the physical domain; for of potential psychic energy we can form no positive representation; we can only say that it is related to actual psychic energy as potential energy is related to actual energy in the physical domain. (Cf. above 8–9.)

In all comparison an urge makes itself felt, even if the finding-out of relations of likeness and difference is not directly set up as a purpose. It is an urge for clarity and distinctness, for the use of the combining capacity that constitutes the innermost essence of consciousness. Often, and especially under the most elementary spiritual conditions, it is a downright urge for self-preservation that leads to thinking. When something that has value for the subsistence of life does not immediately fall to our lot, we must make it a purpose and strive to find the means for its realization. In the relation between end and means, man meets for the first time with necessity: if he wills the end, he must also will the means. The urge for the end begets the urge for the means. Will and thought here go together. This meeting has not merely practical significance. Practically regarded, it is a roundabout way that man here comes to make; but this roundabout way points to a definite unavoidable connection that can become the object of an independent interest, a joy of cognition. In this way the intellectual interest can, through displacement, develop out of the practical one (Cf. 6).

When it is a matter of determining the relation between two objects or of finding an object that stands in a definite relation to a given object, several possibilities will often make themselves felt. Several different possible relations or objects emerge, and a choice must then be made, which is determined by which of them best satisfies the definite conditions. The choice is determined by a comparison between what is required and what offers itself. If the two objects are given, the point is the fixing of the relation that takes place between them; if only the one object and the relation to the sought object are given, the point is the fixing of which object stands in the definite relation to the given object.

As far as possible, comparison leads to an Either—Or, so that there is a choice between two possibilities. The other possibilities are rejected by a preliminary comparison, or they show themselves able to be subsumed under one of the two main possibilities. We stand here before a fundamental peculiarity of our thinking. This moves forward through two-membered relations. A predicate must either be ascribed or not ascribed to a subject. No concept can at once contain and exclude a certain mark. A grounding of this peculiarity of our thinking cannot be given. It is evidently bound up with the psychological fact that attention, during reflection, can be concentrated only on one

element at a time, and that comparison takes place through the passage from one element to another. *George Boole* has called the logical principle springing from this necessity *the Law of Duality*. Before it even the most pronounced practical interest must bow when it wishes to take reflection into its service.

Boole shows in an interesting way the purely factual character of the said feature of our thinking. Starting from the fact that the combination of a concept with itself has no logical significance, he finds the nature of our thinking expressed in the equation $x^2 = x$, which is satisfied only by $x = 1$ or $x = 0$. Here 1 means All, 0 Nothing. The derived equation $x(1 - x) = 0$ expresses the Law of Duality. A combination that should at once contain a concept and its negation is invalid. But if now the fundamental equation had not been of the second, but of the third degree, our thought-activity would be otherwise constituted than it in fact is. We would then not work our way forward through two-membered relations, holding together two elements at a time and affirming or rejecting their likeness (or difference), but through three-membered relations, trichotomies instead of dichotomies—a procedure of which we can form no real representation. Boole by no means wishes, by this presentation, to teach that the fundamental method of our thinking is accidental or arbitrary, but only that it must be taken as a simple fact²⁷.

Already earlier *Salomon Maimon* had shown that, on account of the unity of our consciousness, what is to be connected by our thought in a definite way can immediately consist only of two members. Every immediate comparison that leads to a definite connection concerns two immediately consecutive members. A mediate connection can take place only after a series of immediate connections; thus a mediate connection comes about between a and d when there has successively taken place a connection of a and b, b and c, c and d²⁸.

By Maimon's and Boole's remarks, the law of the excluded middle between two mutually negating assumptions—already set up by *Aristotle*—is illuminated with respect to its connection with the whole nature of our reflection.

23. Besides the object there are, as already said several times, in all reflection two moments that stand in a constant relation of reciprocal action: the form of the thought-work and the motive that awakens reflection and drives it forward.

A motive or an interest that could not be satisfied by the laws and forms that human thinking must by its nature follow will in the long run be without intellectual significance. Such a motive can perhaps become fruitful for imagination, for the capacity to form images and symbols, but it would be a misunderstanding to attempt to force thought

²⁷*Boole: An investigation of the laws of thought*. London 1854. pp. 48–50.

²⁸*S. Maimon: Versuch einer neuen Logik oder Theorie des Denkens*. Berlin 1794. p. 73 f.

onto paths it cannot by its nature walk. Fortunately, “reason” cannot be taken captive forever; it will with inner necessity burst its prison, and its captivity will not be of benefit to the interests that were to be served.

The laws and forms of reflection are the same, whether the thought-work takes place involuntarily or voluntarily, half unconsciously or with full clarity about goal and ways. These laws and forms operate, as we have seen, in analogous forms already in sensation, in memory and imagination, and in association of representations. That the thought-work, the comparison, takes place “with will” entails merely greater definiteness and firmness in the synthesis and greater clarity in the transitions from member to member. However significant the transition from the involuntary to the voluntary can also be, it nonetheless does not mean that a wholly new “capacity” should come into force.

One of the older psychologists who has laid most weight on the will-element in thinking, *Charles Bonnet*, has stated the proposition: L’attention est la mère du génie. But however decisive concentration in a single direction is, it nonetheless gives only the introduction, the impulse. No less important is it that, as a consequence of this impulse, a series of intuitions and associations is involuntarily released, which can then become objects for reflection. In the content of consciousness thus emerging, new elements and new connections can come forth that show themselves able to pass the purgatory of reflection, and genius often consists precisely in the richness and depth of the involuntary intuitions and associations. *Kepler* and *Fechner* are examples of inquirers whose guiding thoughts first came forth in the form of imaginative combinations, which through critical testing led to significant results. Such minds unite the freedom and freshness of the involuntary with the exactness of attention and comparison. Genius can finally also express itself in the indefatigability and carefulness with which the working-over of the objects at hand takes place. Regarded from this side, *Buffon* was right when he defined genius as great patience.

We easily come, in our psychological theories, to emphasize the difference between the involuntary and the voluntary more strongly than close observation warrants. The transition itself from involuntary to voluntary comparison of course takes place involuntarily: the representation of a goal that is the condition for voluntary striving—the one that *Dante* called “the will’s first thought”—must itself first arise by reason of involuntary association. Here can lie a significant turning-point in the life of thought and in the whole development of the mental life; but it does not justify assuming a break in the continuity.

C. Judging

- a. The Formation of Judgment**
- b. Obstacles to the Formation of Judgment**
- c. Subject and Predicate**
- d. The Validity of Judgment**

II. The History of Thought

A. Animism, Platonism, and Positivism

B. Ancient and Modern Thinking

III. The Forms of Thought (The Categories)

A. The History and Method of the Doctrine of Categories

B. Fundamental Categories

C. Formal Categories

a. Identity

b. Quality-Relations

1) Time

2) Number

3) Degree

4) Place

c. Negation

d. Rationality

D. Real Categories

a. Causality

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E. Ideal Categories

1) Formal Differences

2) Real Differences

IV. The Tasks of Thought (Problems)

Introduction

A. Cognition

a. Forms and Principles

b. Forms and Objects

α. Quality and Quantity

β. Succession and Causality

γ. Subject and Object

B. World-View

a. Science and World-View

b. Unity and Multiplicity

c. Spirit and Matter

d. Subsistence and Development

C. Valuation

a. The Ethical Problem

α. *Ethical Work.* (Formal Ethics)

β. *The Rationality of Ethical Valuation.* (Real Ethics)

b. The Religious Problem

α. *The Psychological Place of Religion*

β. *The Historical and the Philosophical Consideration*